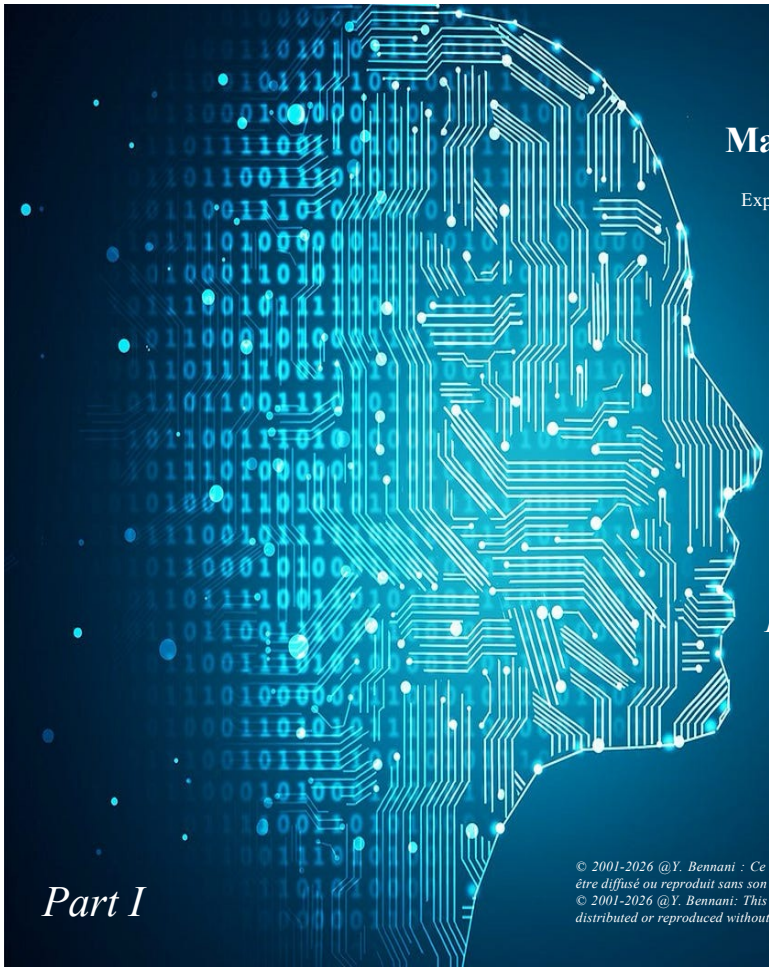


Master of Science in Informatics

Exploration Informatique des Données et Décisionnel (EID^2)
Science des Données (WISD & MASD)
Mathématiques des Données (MATD)

Artificial Neural Networks

From Perceptron to Deep Learning

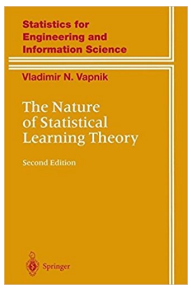
Younès BENNANI
Full Professor

Part I

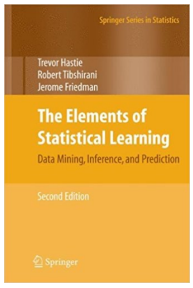
© 2001-2026 @Y. Bennani : Ce document est la propriété de Younès Bennani, Professeur à l'USPN. Il ne peut être diffusé ou reproduit sans son autorisation écrite (younes.bennani@sorbonne-paris-nord.fr).
© 2001-2026 @Y. Bennani: This document is the property of Younès Bennani, Professor at USPN. It may not be distributed or reproduced without his written permission (younes.bennani@sorbonne-paris-nord.fr).

UNIVERSITÉ SORBONNE PARIS NORD | Institut GALILÉE Université Sorbonne Paris Nord

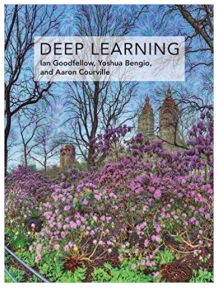
BOOKS



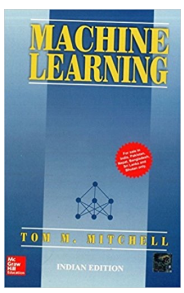
The Nature Of Statistical Learning Theory
by *Vladimir Vapnik*
Second edition – Springer – 2011



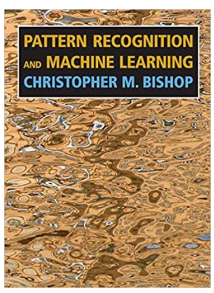
The Elements of Statistical Learning
by *Trevor Hastie, Robert Tibshirani, Jerome Friedman*
Second Edition (Springer Series in Statistics) Hardcover – 2017



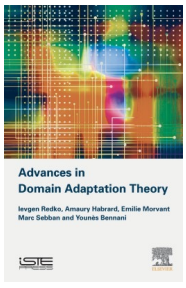
Deep Learning
by *Ian Goodfellow, Yoshua Bengio, Aaron Courville*
Adaptive Computation and Machine Learning series - Hardcover – 2017



Machine Learning
by *Tom M. Mitchell*
Indian Edition – 2017

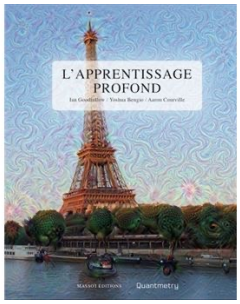


Pattern Recognition and Machine Learning
by *Christopher M. Bishop*
Information Science and Statistics - Hardcover – 2010

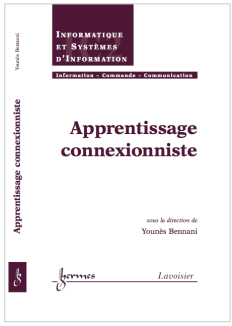


Advances in Domain Adaptation Theory
By *Ievgen Redko, Emilie Morvant, Amaury Habrard, Marc Sebban, Younès Bennani*
Elsevier 2019, ISBN: 9781785482366

BOOKS

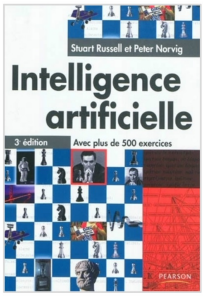


L'apprentissage profond
By *Yoshua Bengio, Aaron Courville, Ian Goodfellow*
Préface Francis Bach - Florent Massot Eds – 2018 -
EAN 979-1097160432 - ISBN 1097160432

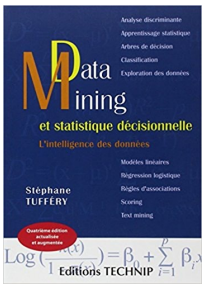


Apprentissage connexionniste
by *Younès Bennani*
Hermès / Lavoisier -2006- Traité IC2 Série
Informatique et systèmes d'information

Apprentissage machine
de la théorie à la pratique
by *Massih-Reza Amini*
Eyrolles – 2015 - EAN13 : 9782212138009



Intelligence artificielle
by *Stuart Russell, Peter Norvig*
Editeur(s) : Pearson, 10/12/2010 (3e édition)
EAN13 : 9782744074554



Data mining et statistique
décisionnelle
by *Stéphane Tufféry*
Editions TECHNIP-2012

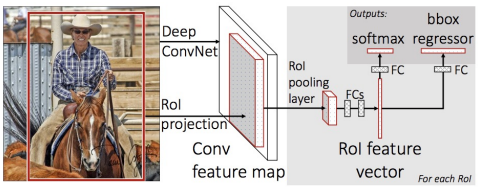
Apprentissage artificiel
Concepts et algorithmes.
by *Antoine Cornuéjols & Laurent Miclet*
Eyrolles – 2010-EAN13 : 9782212124712

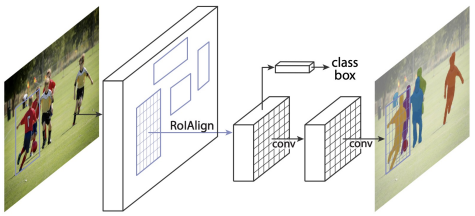
Practical work (lab work)

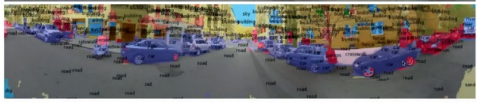
PyTorch

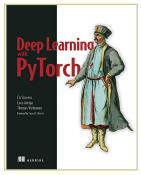
An open source machine learning framework that accelerates the path from research prototyping to production deployment.

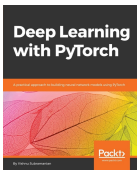
https://pytorch.org

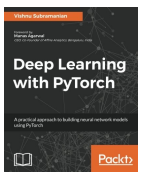


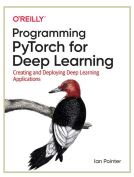


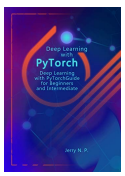












Course materials

Interactive Teaching Space

https://bennani-meziane.com/

Teaching/Deep Learning (DL)

Password: epi-m2-info-dl

Contents

Part I

- Single-layer Neural Networks * Multi-layer perceptron * Radial basis functions * Error functions * Parameter optimization algorithms * Gradient descent optimization * Error Backpropagation * Evaluation of error-function derivatives * Efficiency of backpropagation * Regularization in Neural Networks * Early stopping

Part II

- Convolutional networks * Pooling * Soft weight sharing * Feedforward Deep Networks * Training Criterion and Regularizer * Universal Approximation Properties and Depth * Feature / Representation Learning * Piecewise Linear Hidden Units * Classical Regularization as Noise Robustness

Part III

- Semi-Supervised Training * Unsupervised Pretraining * Dropout * Unsupervised and Transfer Learning * Domain Adaptation * Auto-Encoders * Regularized Auto-Encoders

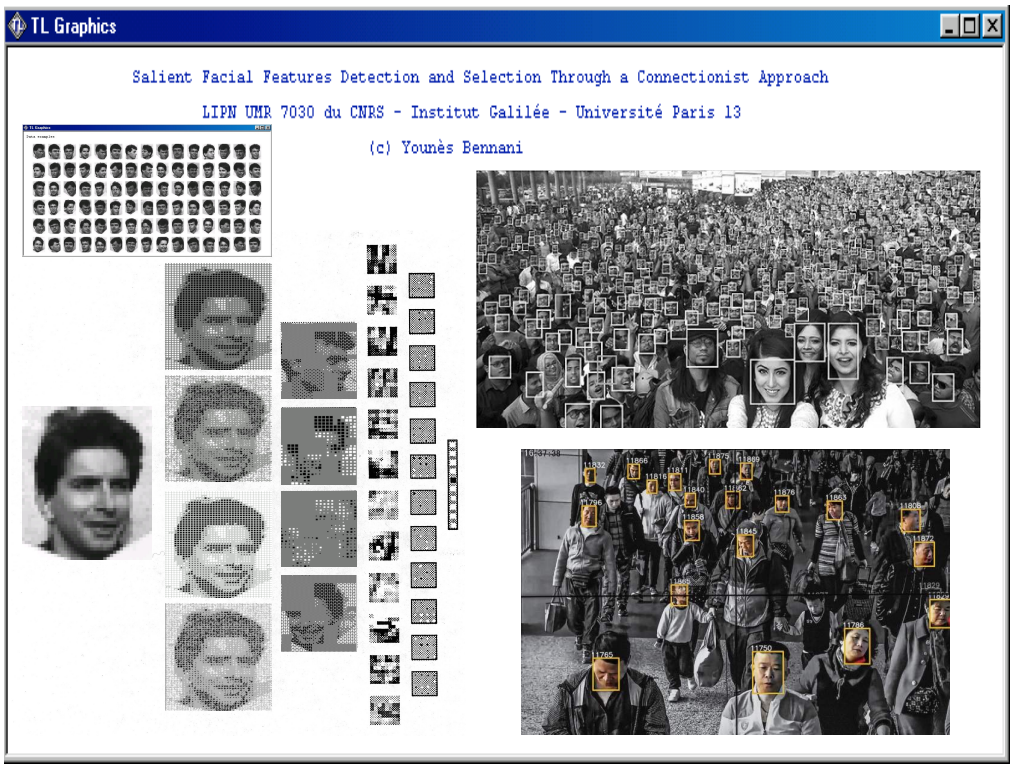
Part IV

- Recurrent Nets * Computing the gradient in a recurrent neural network * Auto-Regressive Networks * Deep generative models * Generative Adversarial Networks * Restricted Boltzmann machines * Deep belief networks * Deep Boltzmann machines * Convolutional Boltzmann machines

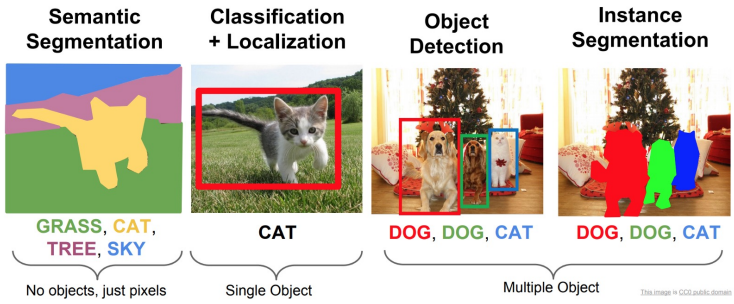
Part V

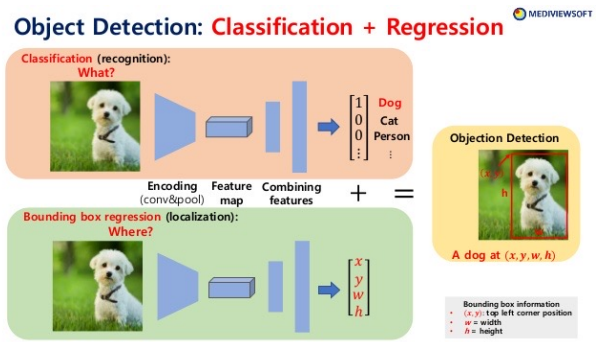
- Attention Mechanisms *Multi-head Attention•Similarity *The architecture of a transformer model.

Artificial Intelligence (AI) can isolate your face from a crowd



AI for the detection and segmentation of objects of interest





AI for crowd segmentation & counting

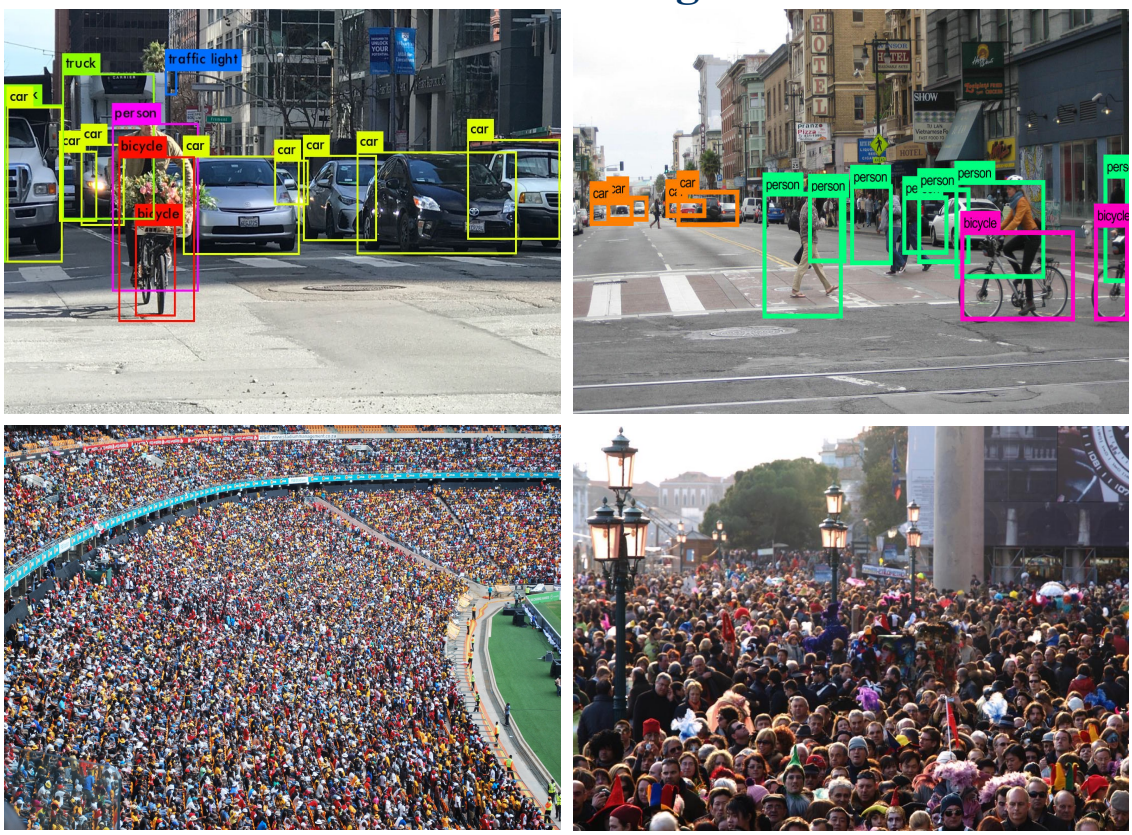
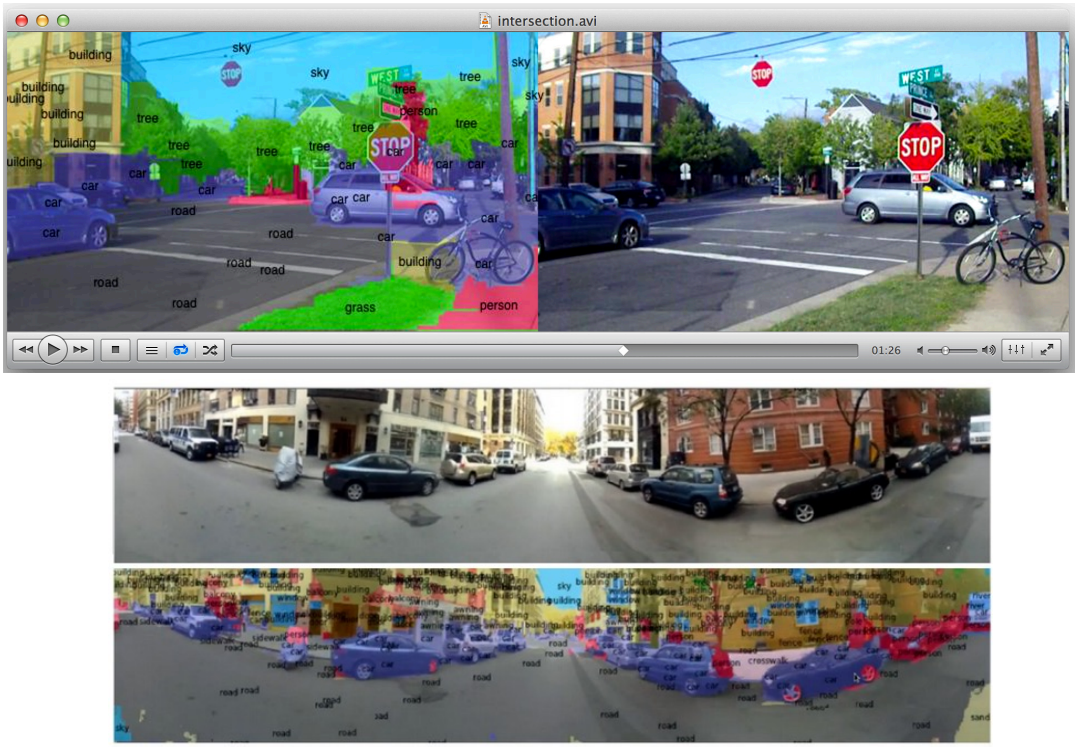
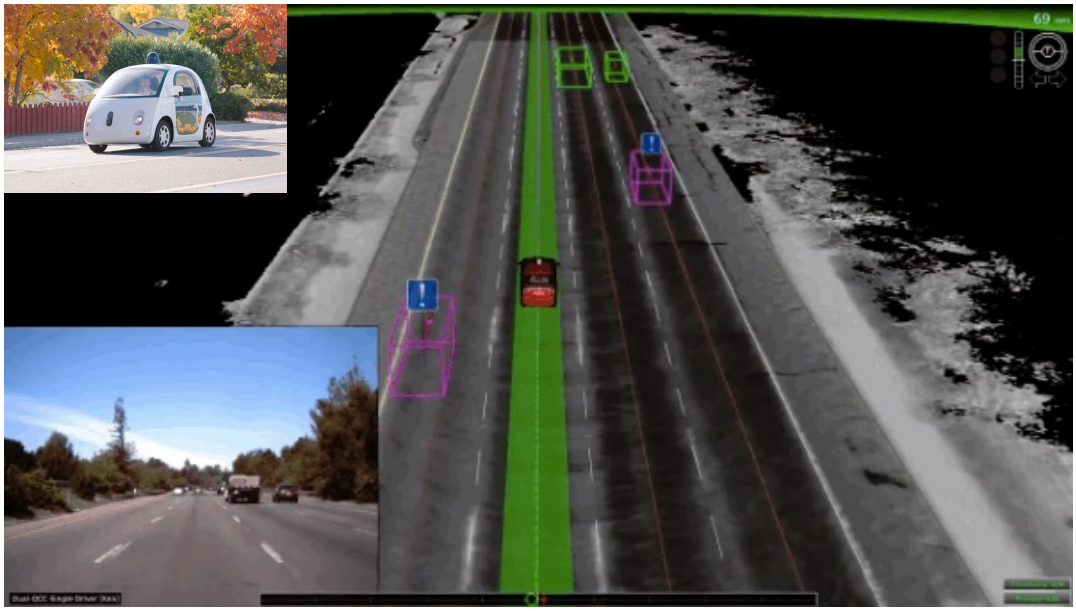


Image segmentation



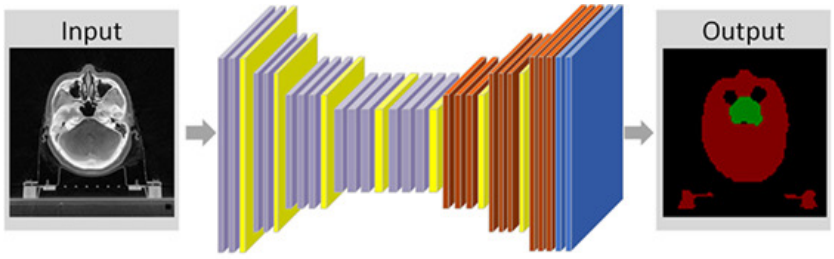
AI triumphs in vision: spectacular results

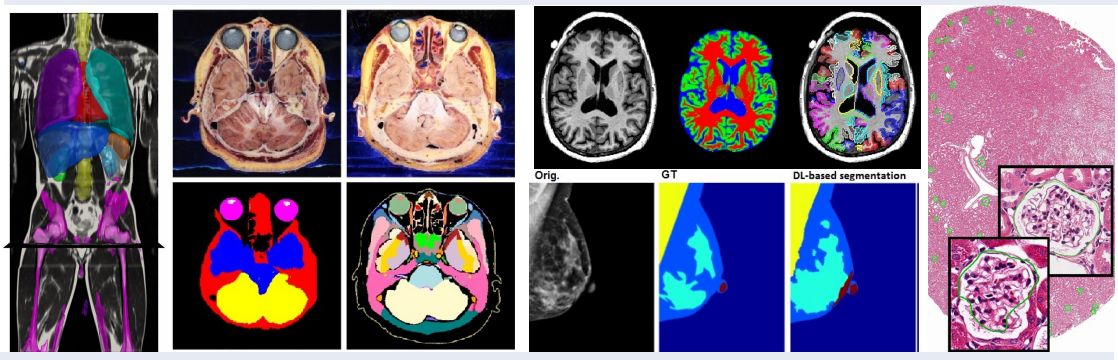
Google Self-Driving Car Project



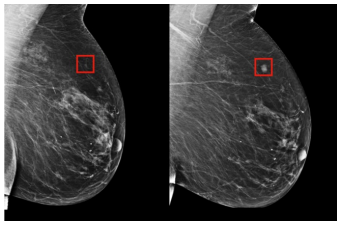
MIT: the "autonomous" car

Diving into medical imaging: the enlightened view of AI

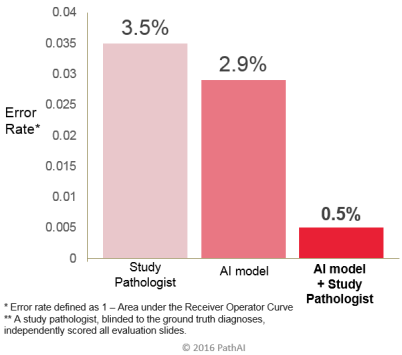




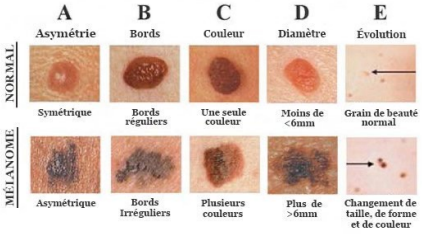
Medical diagnosis: spectacular results!

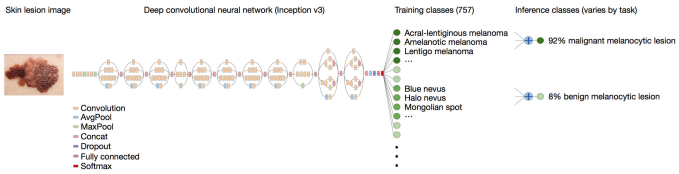


(AI + Pathologist) > Pathologist



Expert level skin cancer classification.
(Nature, 2017)



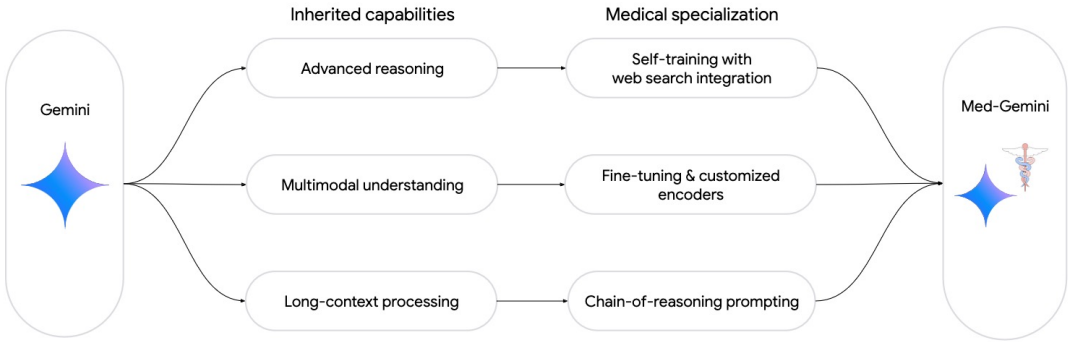


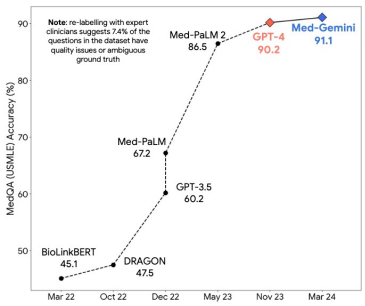
« ***Dermatologist-level Classification of Skin Cancer with Deep Neural Networks*** »
Andre Esteva, Brett Kuprel, Rob Novoa, Justin Ko, Susan Swetter, Helen Blau, Sebastian Thrun - Nature, 2017

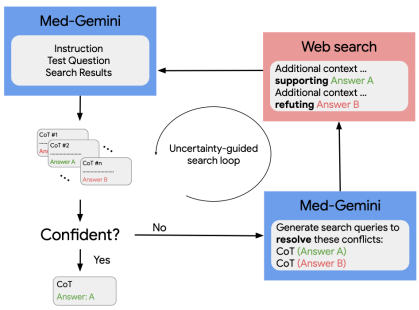
Large Language Models (LLMs): inspired medical storytellers

Google DeepMind

Med-Gemini Development







K. Saab, T. Tu, W-H. Weng, et al. (2024), « Capabilities of Gemini Models in Medicine », arXiv:2404.18416.

Stories in pulmonology: a breath of innovation

Hello, I am a primary care physician and this is a chest x-ray for a patient of mine. The formal radiology report is still pending and I would like some help to understand the x-ray. Please write a radiology report for me:

FINDINGS: The heart is at the upper limits of normal size. The mediastinal and hilar contours appear within normal limits. The lungs appear clear. There are no pleural effusions or pneumothorax. Mild degenerative changes are noted along the lower thoracic spine. IMPRESSION: No evidence of acute disease.

What evidence do you see for degenerative changes?

Mild degenerative changes are noted along the lower thoracic spine. This can be seen by the slight narrowing of the intervertebral disc spaces and the presence of small osteophytes (bone spurs) along the anterior margins of the vertebral bodies.

My patient has a history of back pain. Might this explain the back pain?

While the degenerative changes seen on the x-ray could potentially contribute to your patient's back pain, it's important to note that correlation doesn't equal causation. Many people have degenerative changes in their spine without experiencing pain, and conversely, back pain can have numerous causes beyond just degenerative changes.

Therefore, while the x-ray findings are worth noting, they don't definitively explain your patient's back pain. A comprehensive evaluation, including a detailed history, physical examination, and potentially additional imaging or tests, would be necessary to determine the exact cause of their pain.

Please let me know if you have any other questions.

To help me communicate all of this to my patient, please rewrite the report without medical terminology:

Chest X-ray Results:

- **Heart:** Normal size.
- **Lungs:** Clear, no signs of infection or fluid buildup.
- **Bones:** Minor wear and tear in the lower spine, common with age.

Overall: No signs of serious problems in the chest. The wear and tear in the lower spine might be related to your back pain, but other factors could also be involved. We'll need to do more tests to figure out the exact cause of your pain.

Thank you. That is useful while I wait for the radiologists report.

You're welcome! I'm glad I could help. If you have any further questions, feel free to ask.

Ground truth report *(not provided to model)*

FINDINGS: The cardiomediastinal and hilar contours are normal. The lungs are well expanded and clear, without focal consolidation, pleural effusion or pneumothorax. Mild degenerative changes are seen in the thoracic spine. IMPRESSION: No acute cardiopulmonary pathology.

Example of a hypothetical multimodal dialogue between diagnosis and assistance with Med-Gemini-M 1.5 in radiology.

Radiologist's comments:

Pros

- "Realistic hypothetical demonstration of interaction with a healthcare provider in a clinical imaging context."
- "Demonstration of high-quality realistic report generation."
- "Showcases general medical knowledge and also provides adequate limitations (i.e., mentioning additional tests needed)."

K. Saab, T. Tu, W-H. Weng, et al. (2024), « Capabilities of Gemini Models in Medicine », arXiv:2404.18416.

Stories in surgery: the digital assistant in the operating theatre

Hello, I am a medical student analyzing this surgical procedure video for my studies. Which surgical procedure is shown in this video?

The surgical procedure shown in this video is a laparoscopic cholecystectomy - the surgical removal of the gallbladder. This is a common procedure and is often performed when gallstones cause pain or complications, such as inflammation of the gallbladder (cholecystitis).

Which tools and how many are being used by the surgeon?

The surgeon in this video clip is using two tools:
Laparoscopic Grasper: A long, thin instrument with a handle on one end and a grasping mechanism on the other, which can be opened and closed to hold tissues, and manipulate organs and other structures.
Hook Electrocautery: A long thin instrument with a handle on one end and a hook-shaped tip on the other. The hook shape allows the tool to dissect, cut and cauterize tissue. Cauterizing tissue helps prevent bleeding during surgery and seals off blood vessels and lymphatic ducts to minimize the risk of leaks.

Which anatomical structures are visible?

The following anatomical structures are visible:

Liver: The large, reddish-brown organ in the upper right portion of the abdomen.
Gallbladder: The small, pear-shaped organ located below the liver. The gallbladder is grasped and retracted by forceps.
Cystic duct: The thin, tube-like structure that connects the gallbladder to the common bile duct.
Common bile duct: The larger duct visible below the cystic duct that carries bile from the liver and gallbladder to the small intestine.
Cystic artery: The small artery that supplies blood to the gallbladder, it is visible alongside the cystic duct.

Which specific surgical phase is shown this video clip?

The specific surgical phase shown in this video clip is the dissection and isolation of the cystic duct and artery. This phase comes after the initial access to the abdominal cavity and visualization of the gallbladder, which can be seen retracted superiorly by forceps, and before the clipping and dividing of the cystic duct and artery.
The surgeon is using a laparoscopic dissector to carefully separate the cystic duct and cystic artery from the surrounding tissue. This is a critical step as it allows the surgeon to clearly identify and isolate these structures before they are clipped and divided and can help to prevent injury to the common bile duct and hepatic artery.

Which steps should the surgeon take to avoid injuring those structures?

To prevent such injuries, the surgeon should consider the following:
Critical View of Safety (CVS): This technique involves dissecting and clearing the Calot's triangle to clearly visualize the cystic duct and artery independently before clipping or cutting to ensure that they are not mistaken for the common bile duct or hepatic artery.
Intraoperative Cholangiogram (IOC): Although not frequently performed now, this procedure involves injecting a contrast dye into the cystic duct to visualize the biliary tree on X-ray. This can help the surgeon to confirm the anatomy of the bile ducts and identify any abnormal structures, ensuring the common bile duct is not clipped.
Fluoroscopy By using fluoroscopy during the surgery to accurately identify the cystic duct and common bile duct using contrast.
Stay within Calot's triangle - The area bordered by the cystic duct, the common hepatic duct, and the cystic artery.
Do you want to learn more about the specific steps?

An example of Med-Gemini-M 1.5's long context capabilities on a surgical video dialogue.

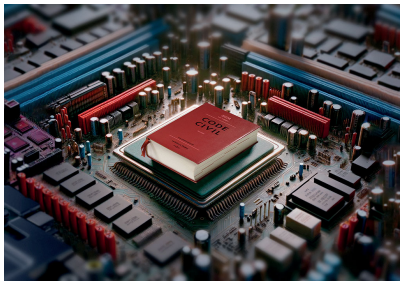
Surgeon's comments:

Pros

- "Impressive understanding of which surgical procedure is being performed, which specific phase in the procedure this video clip is taken from, which tools are being used, and which anatomical structures are visible."
- "Correctly identifies the risks in the current video clip, which is injuring the common bile duct, and correctly suggests taking the safety measures, highlighting the Critical View of Safety method first."

K. Saab, T. Tu, W-H. Weng, et al. (2024), « Capabilities of Gemini Models in Medicine », arXiv:2404.18416.

AI in the field of law and justice



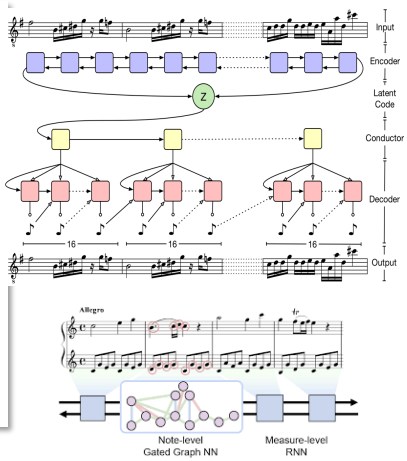
Livre blanc Lefebvre Dalloz : **Révolutionner le secteur juridique avec l'intelligence artificielle**

- Use of generative AI to produce automated legal summaries, analyse case law and suggest decisions.
- Advantages: time savings, consistency in case analysis, reduction in human error.
- Risks: bias in the legal data used to train the models, lack of contextual understanding of the subtleties of laws and individual cases.
- Ethical debates: how far should AI be allowed to intervene in judicial decision-making?

Tool	Country	Year	Typeof tool	Website
GenIA-L	France	2023	Legal research	www.lefebvre-dalloz.fr/
Ordalie	France	2022	Contract drafting	www.ordalie.fr/
Jiminy AI	France	2021	Drafting of documents	www.jiminy-ai.com/
WiseTax	France	2021	Tax	www.wisetax.fr/
Smart Lawyer	France	2019	Legal managemnt	www.smartlawyer.fr/
Lexis Nexis	France	2018	Advanced research	www.lexisnexis.fr/
Case Law Analytics	France	2017	Decision support	www.caselawanalytics.com/
Josef	Australie	2017	Automation of legal documents	www.joseflegal.com/
BlackBoiler	États-Unis	2016	Automation of legal documents	www.blackboiler.com/
Doctrine	France	2016	Legal reserach	www.doctrine.fr/
Blue J	Canada	2015	Decision support	www.bluej.com/
LegalUp	Israël	2015	Automation of legal documents	www.legalup.me/
Luminance	Grande-Bretagne	2015	Automation of legal documents	www.luminance.com/
Predictice	France	2015	Legal research	www.predictice.com/
LawGeex/Superlegal	Israël	2014	Automation of legal documents	www.lawgeex.com/
CaseText	États-Unis	2013	Legal research	www.casetext.com/
CanLII	Canada	2001	Legal research	www.canlii.org/

The AI invites itself into the world of art ...

A work painted by an AI was sold at auction for over $430,000 in New York.

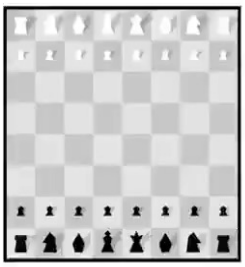


115 years after AI completed a symphony by Czech composer **Antonín Dvořák**

AI "plays" better than the great champions

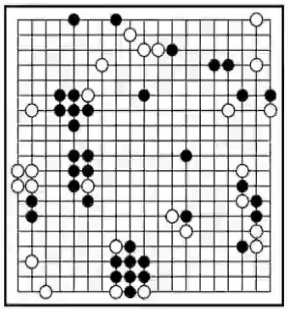
- ✓ Chess: the famous defeat in 1997 of the then world chess champion Garry Kasparov against an AI designed by IBM: **Deep Blue**.

Chess

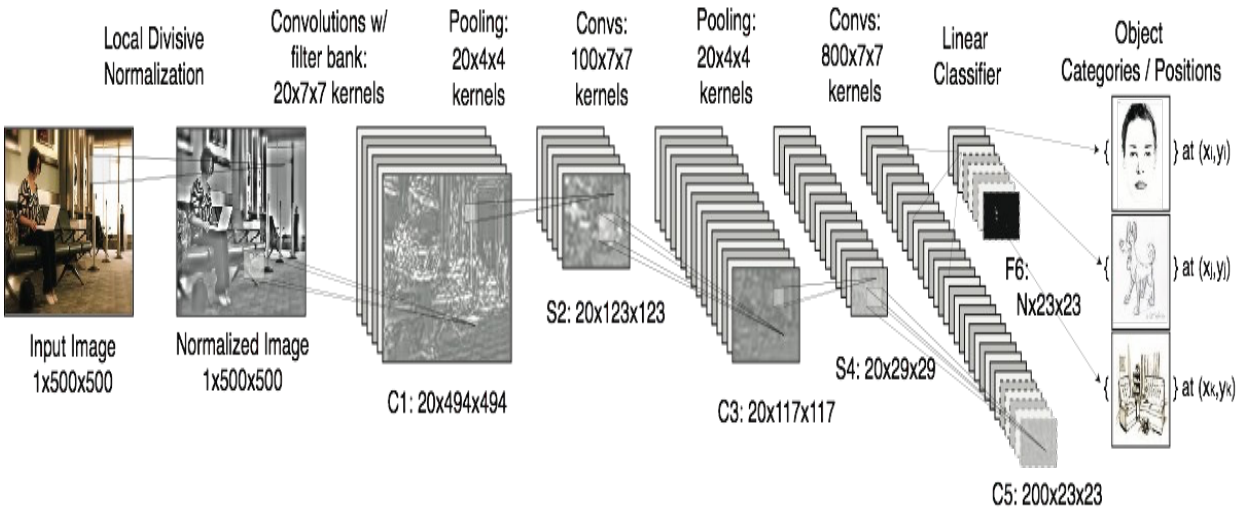


- ✓ In 2017, **AlphaGo Zero** beat all the champions!
  - in **3 hours**, AlphaGo Zero learned the fundamentals of the game of Go
  - **after 21 days**, it equalled AlphaGo Master, which had beaten world champion Ke Jie

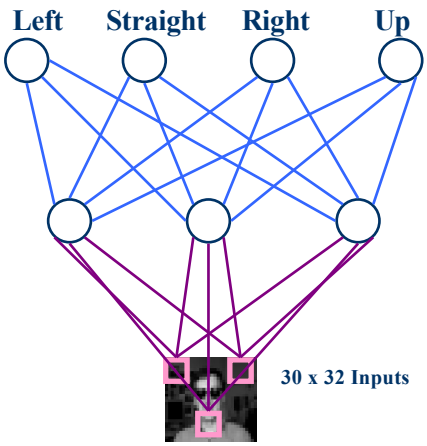
Go

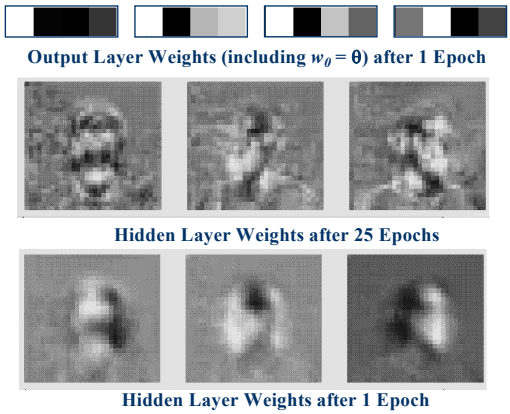


Categories/Positions in an image

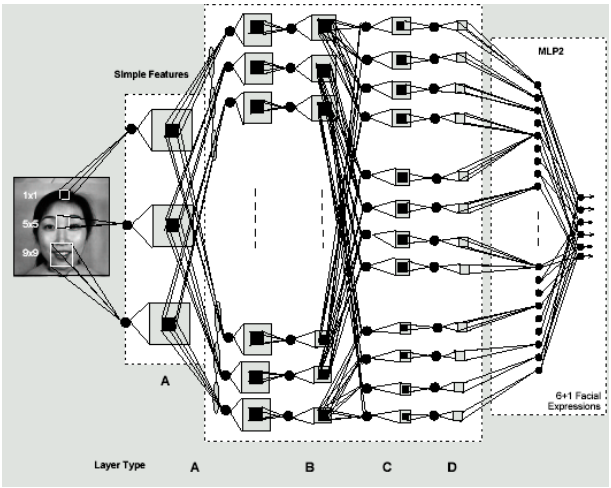


Position of the face in a picture





Expression of the face in a picture

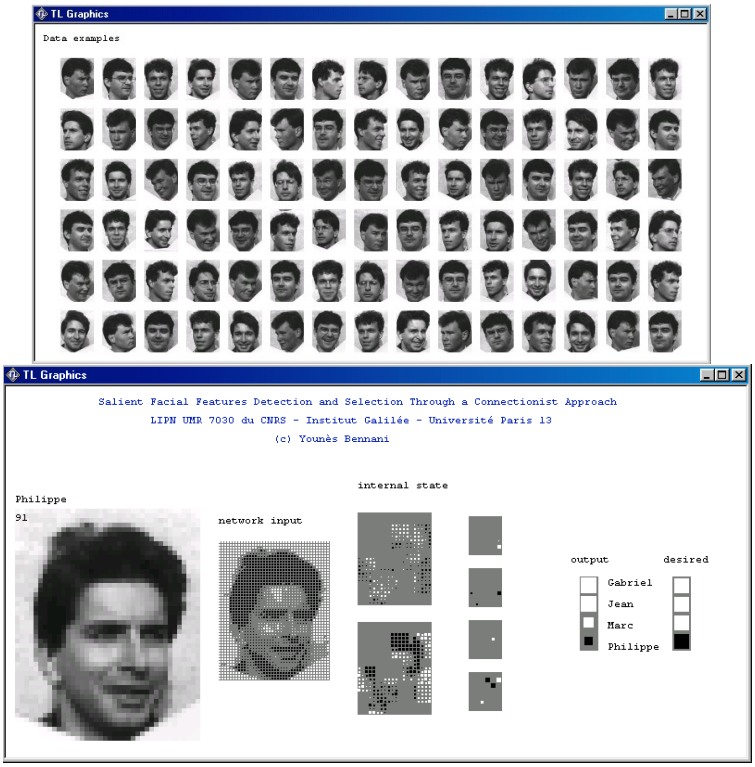


210 images (246x256 : TIFF)

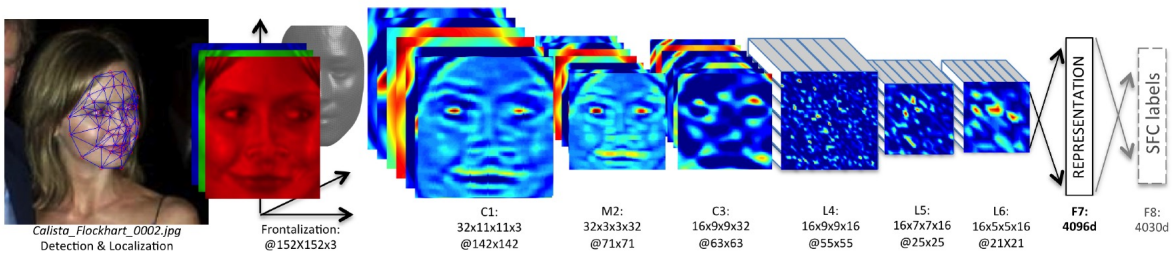
10 femmes japonaises

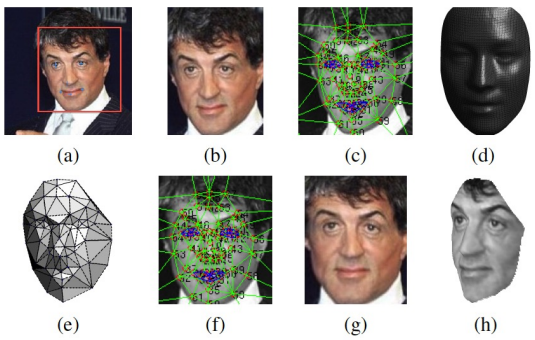
6 expressions + 1 position neutre

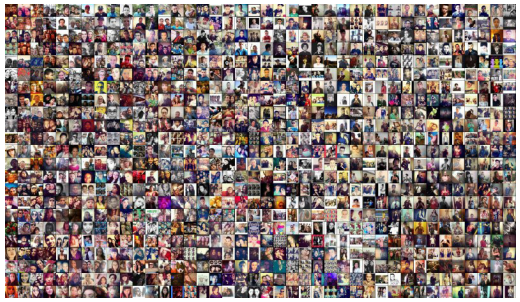
Facial Recognition



"DeepFace", Facebook's new facial recognition system





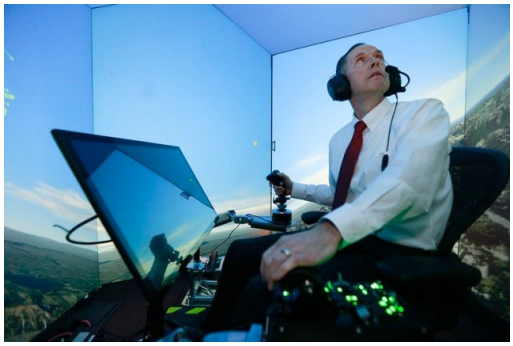


DeepFace: Closing the Gap to Human-Level Performance in Face Verification

Yaniv Taigman Ming Yang Marc'Aurelio Ranzato Lior Wolf
Facebook AI Group
Menlo Park, CA, USA
{yaniv, mingyang, ranzato}@fb.com
Tel Aviv University
Tel Aviv, Israel
wolf@cs.tau.ac.il

Unmanned aeroplanes soon? Google Planes!

Google Planes



Once again, artificial intelligence is knocking down a big name in its field: General **Gene Lee**, trainer of pilots for the **US Air Force**. AI has won all the rounds of the simulation.

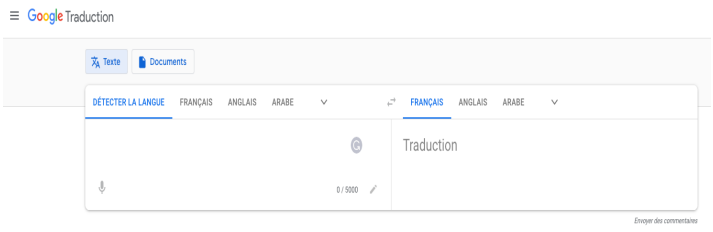
"J'ai été surpris par sa réactivité. Alpha semblait au fait de toutes mes intentions et réagissait instantanément à mes changements de vol et mes déploiements de missile"

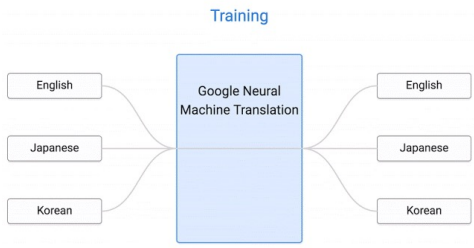
Gene Lee
Juillet 2016

"I was surprised by its responsiveness. Alpha seemed to be aware of all my intentions and reacted instantly to my flight changes and missile deployments".

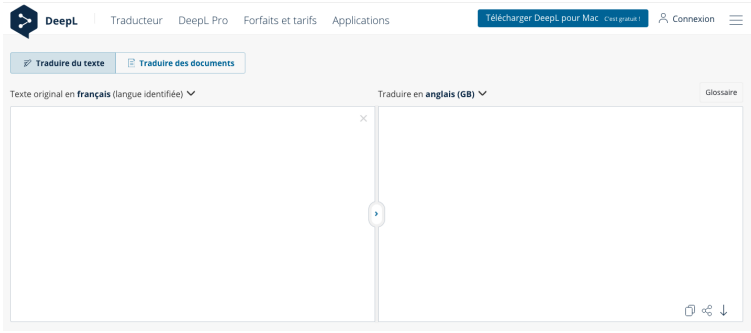
Gene Lee
July 2016

Eliminating language barriers Worldwide by using AI





DeepL is a German company that has set itself the goal of eliminating language barriers worldwide by using artificial intelligence. (www.DeepL.com)



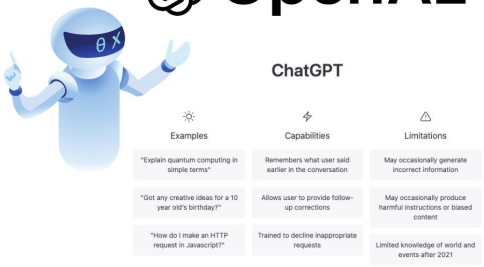
ChatGPT:
Chat Generative Pre-trained Transformer

deepseek

Mistral AI

ChatGPT

OpenAI



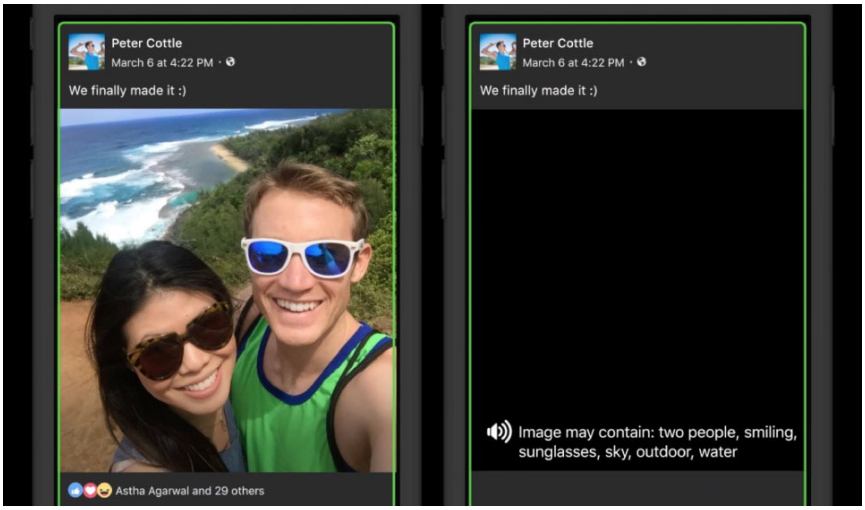
Write a short text to welcome the students of the EID2 master in the course on "Deep Learning" given by professor Younès Bennani.

Welcome to the Deep Learning course, students of the EID2 master program! We are excited to have you with us and to embark on this journey of learning together. Professor Younès Bennani is a highly respected expert in the field of Deep Learning, and we are fortunate to have him as our instructor for this course.

Over the next few months, we will be diving into the exciting world of Deep Learning and exploring its many applications. You will learn about the fundamental concepts and techniques that underlie this powerful approach to machine learning, and you will have the opportunity to apply what you have learned to a variety of real-world problems.

We hope that this course will be both challenging and rewarding for you, and that you will emerge from it with a strong foundation in Deep Learning that will serve you well in your future studies and career. So let's get started! We are looking forward to an exciting and productive semester together.

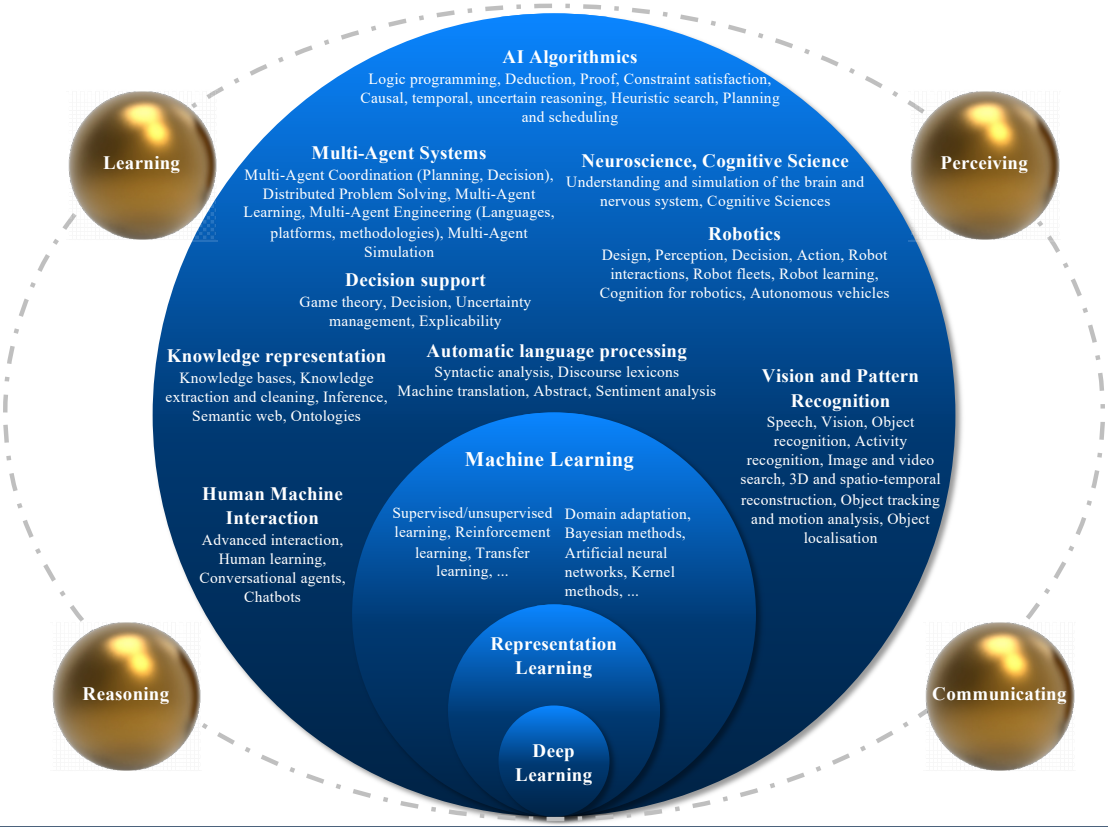
Assistance for the blind



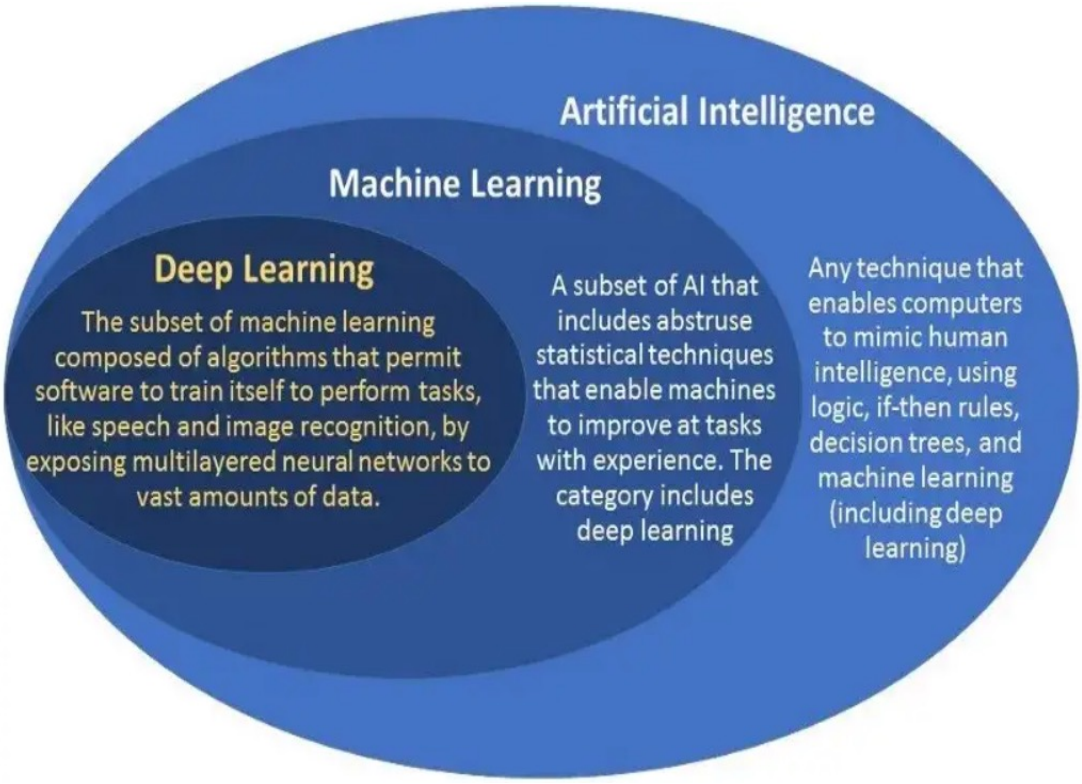
DeepDream: Google's psychedelic dream machine

DeepDream: Google's psychedelic dream machine

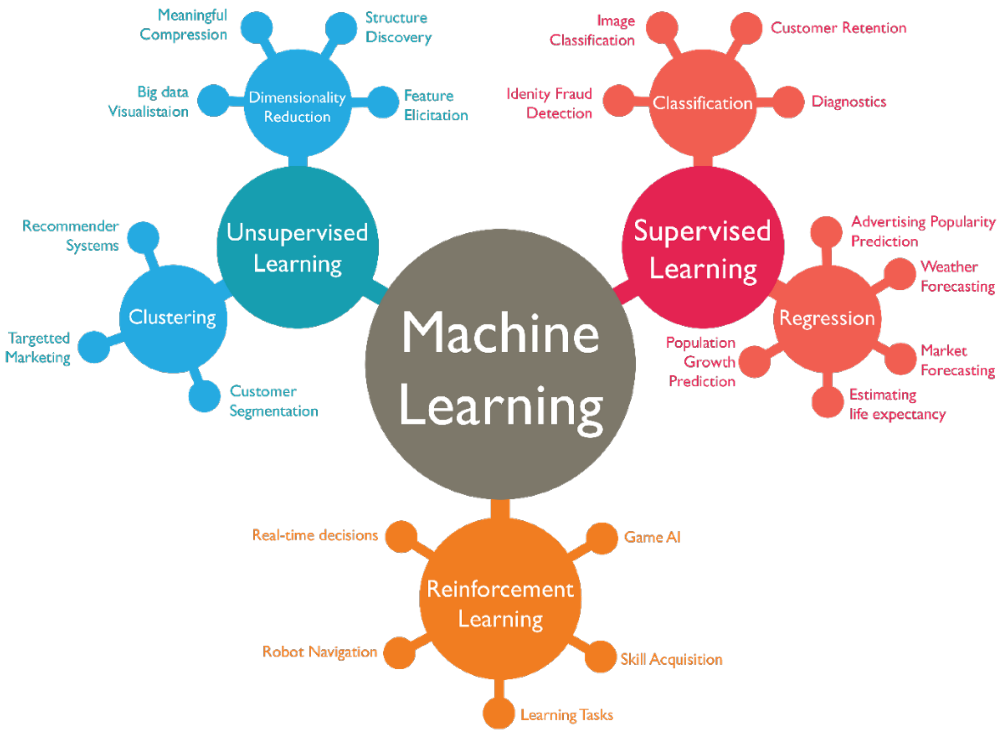
AI & Machine Learning

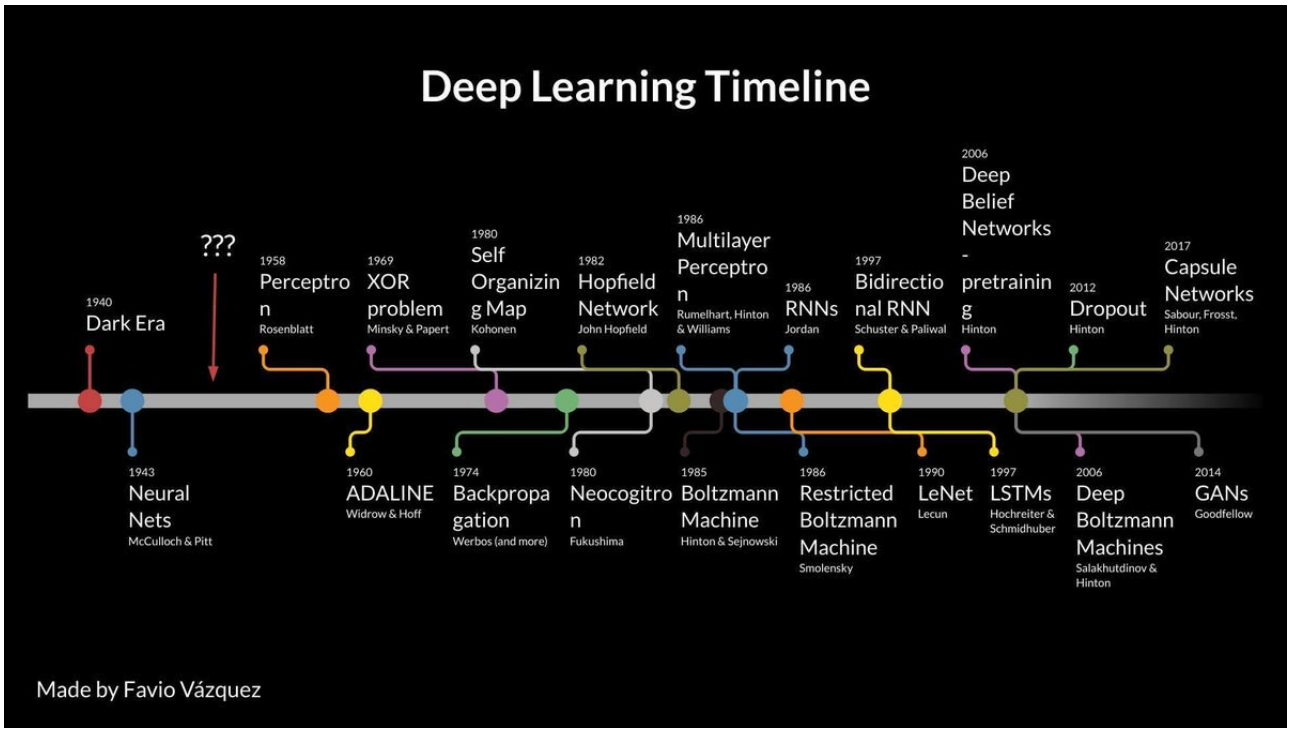


AI & Machine Learning



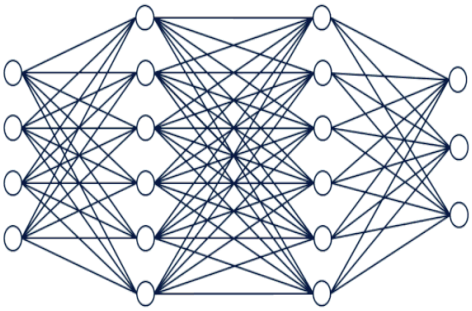
Machine Learning paradigms taxonomy





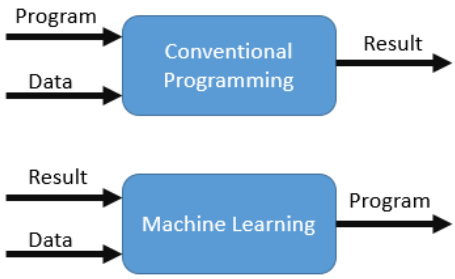
What is Machine Learning?

Input space **X**

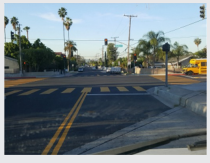
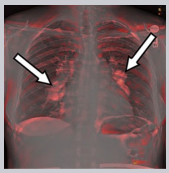


Smooth morphing from **X** to **Y** expressed as a series of simple geometric transformations (layers)

Output space **Y**



What is Machine Learning?

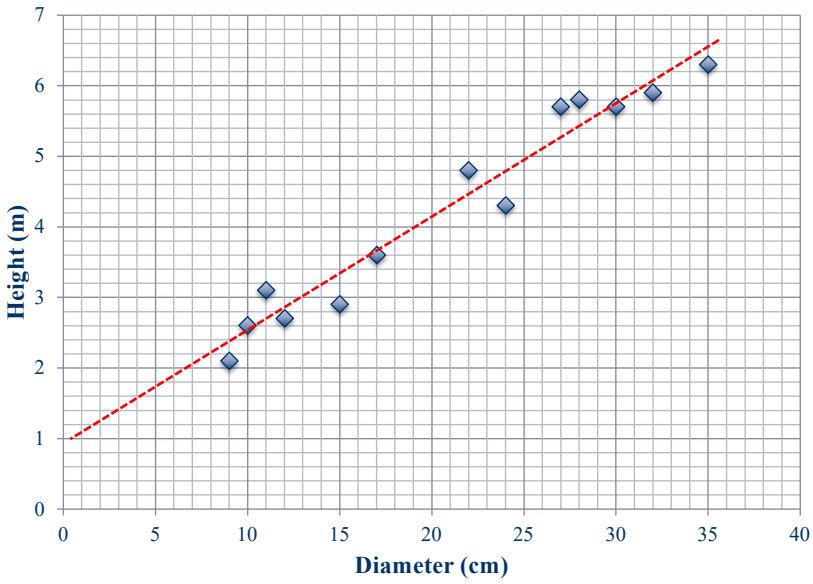
Task	Input	Output
Digit classification		5 7 8
Next-Word-Prediction	Looking forward to your …	replay
Playing GO		Next move
Segmentation		
Diagnosis		

What is Machine Learning?

Diameter (cm)	Height (m)
12	2,7
19	4,7
15	3,9
11	3,1
23	5,9
27	6,7
32	8,2
10	2,6
28	6,3
35	9,2
20	6,4
9	2,1
17	3,6

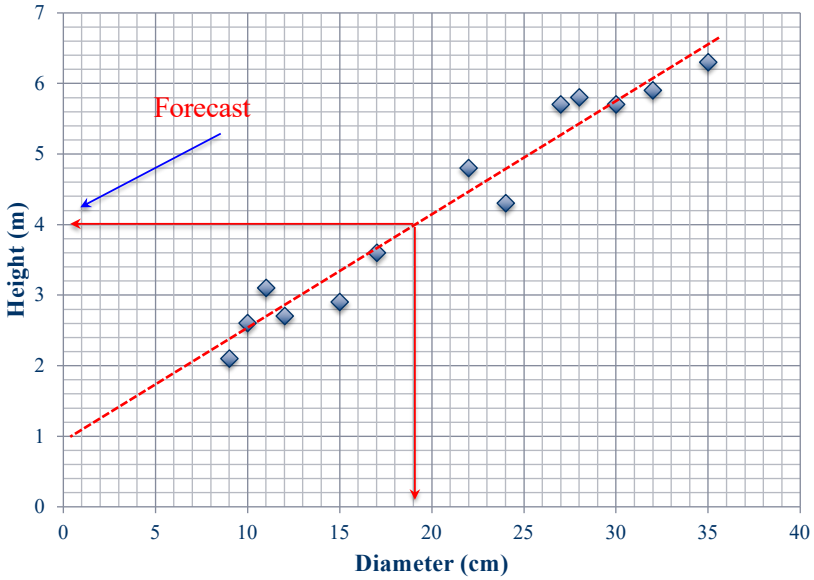
What is Machine Learning?

Diameter (cm)	Height (m)
12	2,7
19	4,7
15	3,9
11	3,1
23	5,9
27	6,7
32	8,2
10	2,6
28	6,3
35	9,2
20	6,4
9	2,1
17	3,6



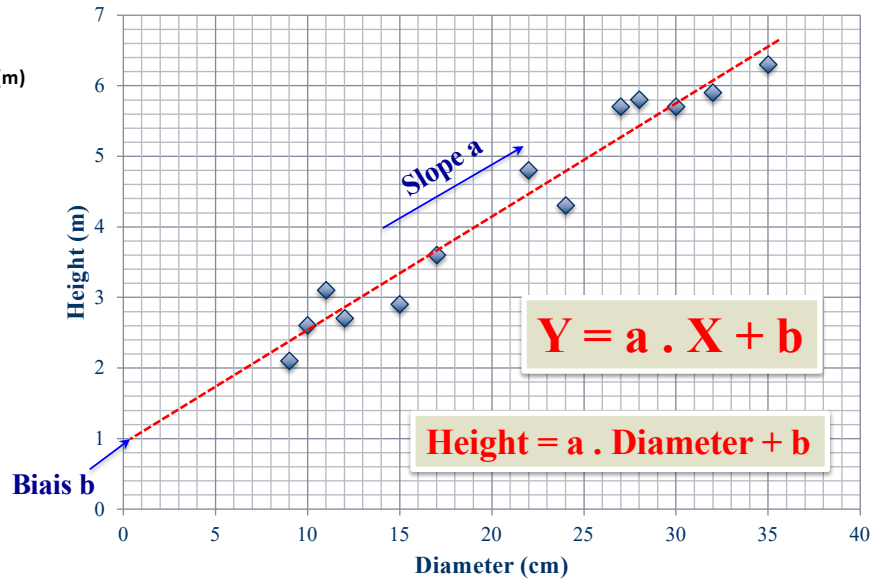
What is Machine Learning?

Diameter (cm)	Height (m)
12	2,7
19	4,7
15	3,9
11	3,1
23	5,9
27	6,7
32	8,2
10	2,6
28	6,3
35	9,2
20	6,4
9	2,1
17	3,6
19	?



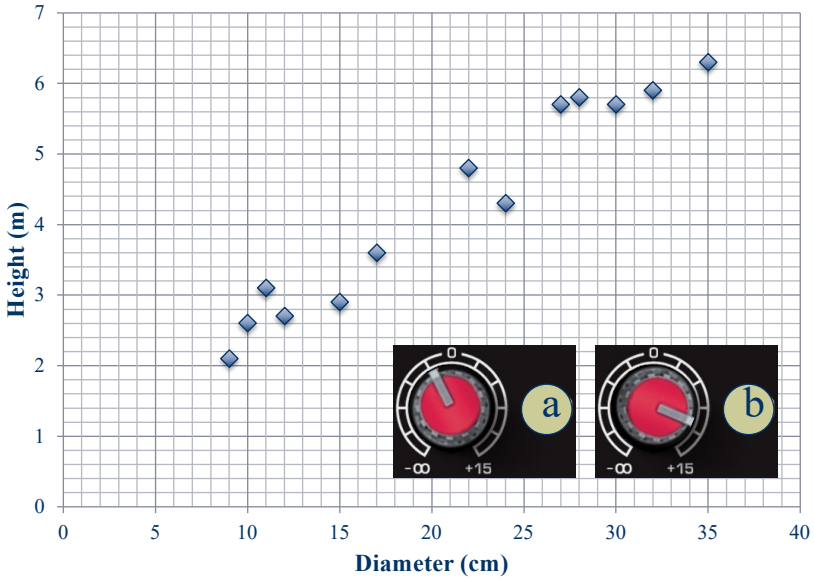
What is Machine Learning?

Diameter (cm)	Height (m)
12	2,7
19	4,7
15	3,9
11	3,1
23	5,9
27	6,7
32	8,2
10	2,6
28	6,3
35	9,2
20	6,4
9	2,1
17	3,6



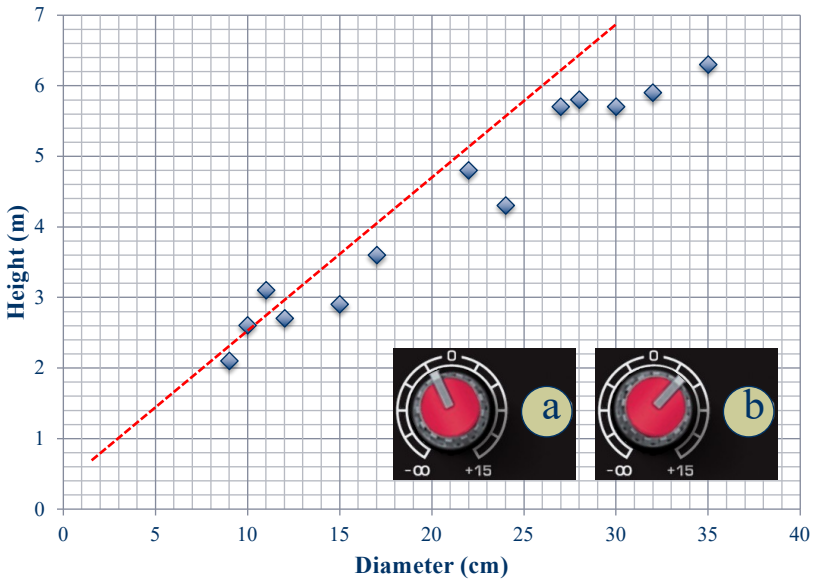
What is Machine Learning?

Diameter (cm)	Height (m)
12	2,7
19	4,7
15	3,9
11	3,1
23	5,9
27	6,7
32	8,2
10	2,6
28	6,3
35	9,2
20	6,4
9	2,1
17	3,6



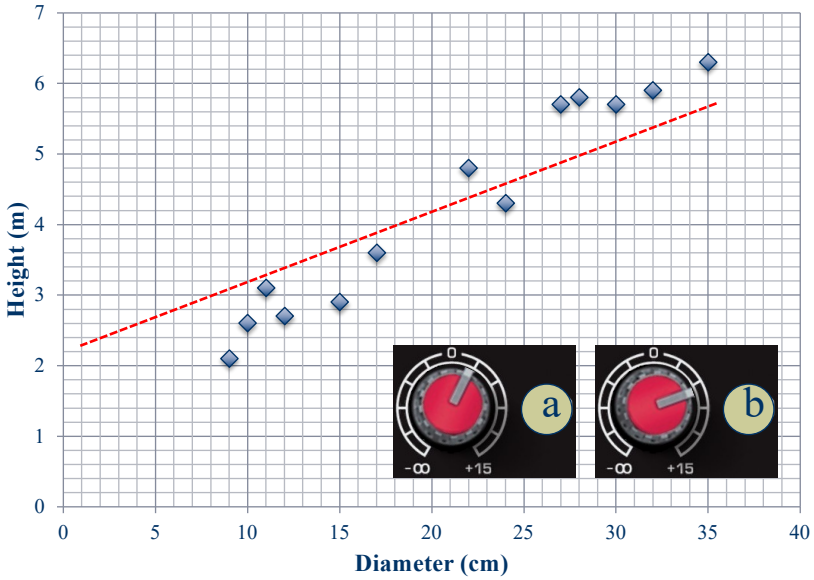
What is Machine Learning?

Diameter (cm)	Height (m)
12	2,7
19	4,7
15	3,9
11	3,1
23	5,9
27	6,7
32	8,2
10	2,6
28	6,3
35	9,2
20	6,4
9	2,1
17	3,6



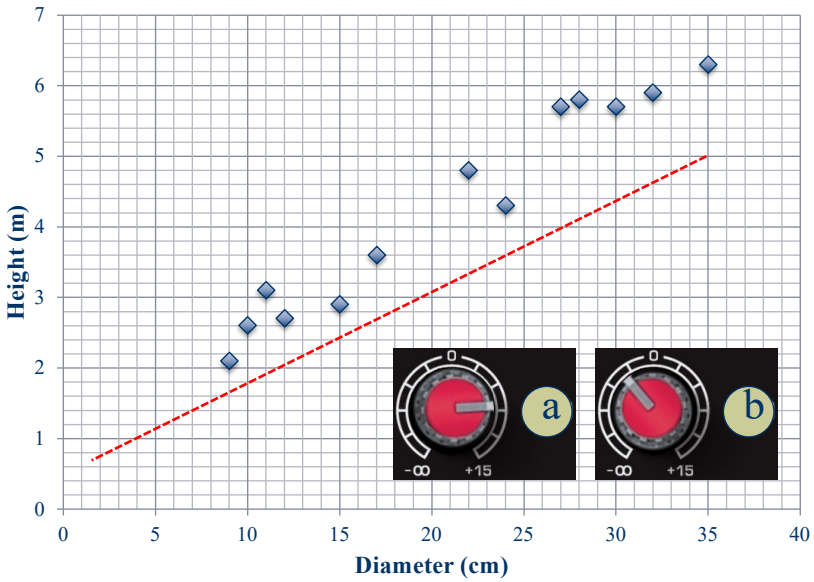
What is Machine Learning?

Diameter (cm)	Height (m)
12	2,7
19	4,7
15	3,9
11	3,1
23	5,9
27	6,7
32	8,2
10	2,6
28	6,3
35	9,2
20	6,4
9	2,1
17	3,6



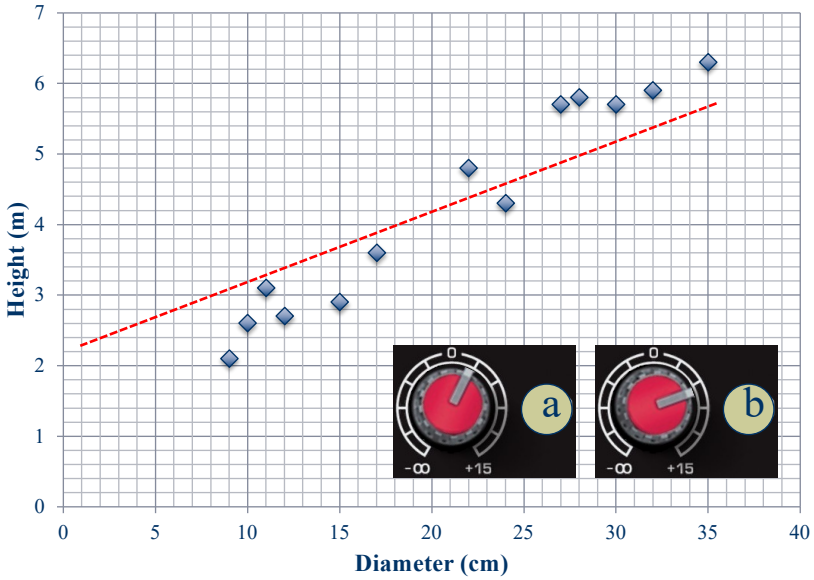
What is Machine Learning?

Diameter (cm)	Height (m)
12	2,7
19	4,7
15	3,9
11	3,1
23	5,9
27	6,7
32	8,2
10	2,6
28	6,3
35	9,2
20	6,4
9	2,1
17	3,6



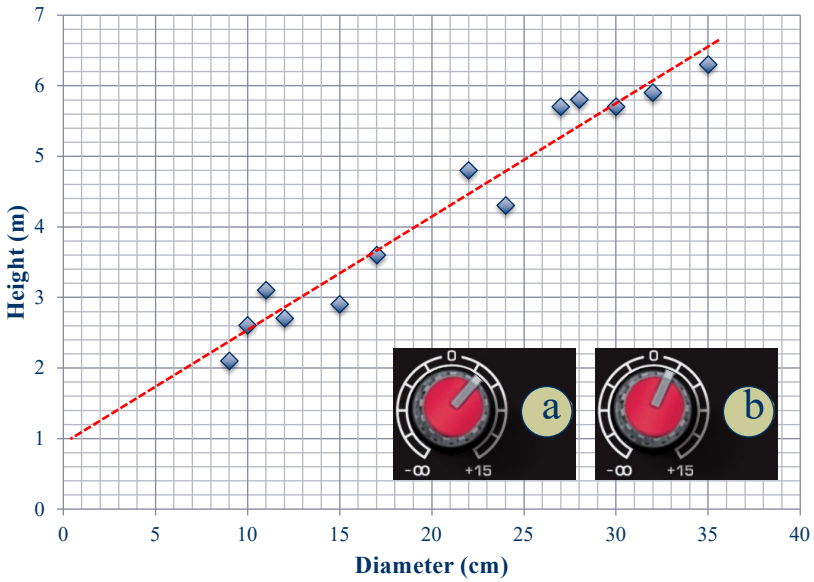
What is Machine Learning?

Diameter (cm)	Height (m)
12	2,7
19	4,7
15	3,9
11	3,1
23	5,9
27	6,7
32	8,2
10	2,6
28	6,3
35	9,2
20	6,4
9	2,1
17	3,6



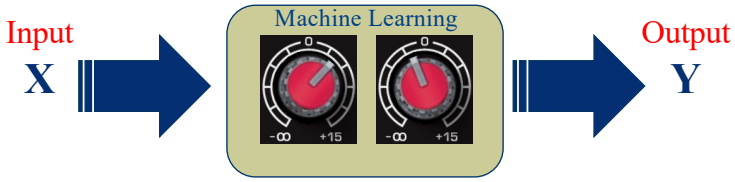
What is Machine Learning?

Diameter (cm)	Height (m)
12	2,7
19	4,7
15	3,9
11	3,1
23	5,9
27	6,7
32	8,2
10	2,6
28	6,3
35	9,2
20	6,4
9	2,1
17	3,6



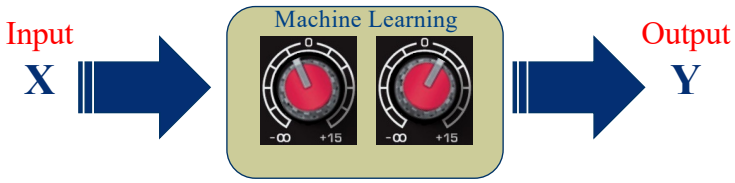
What is Machine Learning?

X	Y
12	2,7
19	4,7
15	3,9
11	3,1
23	5,9
27	6,7
32	8,2
10	2,6
28	6,3
35	9,2
20	6,4
9	2,1
17	3,6



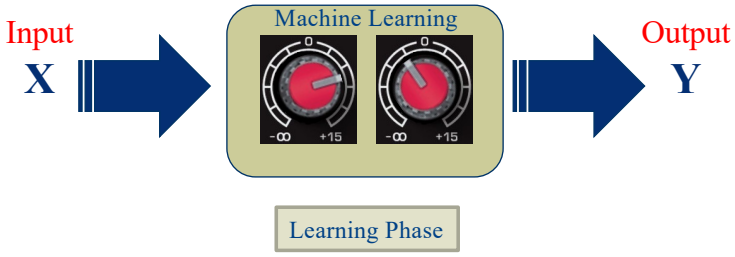
What is Machine Learning?

X	Y
12	2,7
19	4,7
15	3,9
11	3,1
23	5,9
27	6,7
32	8,2
10	2,6
28	6,3
35	9,2
20	6,4
9	2,1
17	3,6



What is Machine Learning?

X	Y
12	2,7
19	4,7
15	3,9
11	3,1
23	5,9
27	6,7
32	8,2
10	2,6
28	6,3
35	9,2
20	6,4
9	2,1
17	3,6



How does AI discover the laws of the land of perfect gases?

Thermodynamics: Law of perfect gases

$$P.V = n.R.T$$

P represents the pressure (in kPa)
V is the volume (in L)
n is the quantity of gas (in mol)
R is the perfect gas constant (8.314 kPa·L/mol·K)
T is the absolute temperature (in K)

Problem:

Given a gas at pressure P and temperature T, what is the quantity n in moles of this gas?

$$n = \frac{P.V}{R.T}$$

How does AI discover the laws of the land of perfect gases?

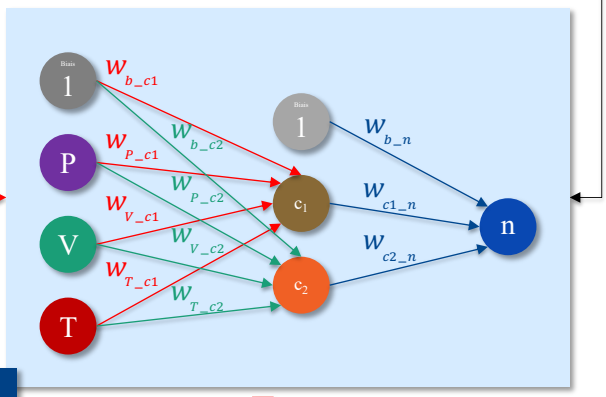
Input | Output

#	Pressur (kPa)	Volume (L)	Temperature (K)	Quantity of gas (mol)
1	100	10	300	4
2	150	15	350	6
3	200	20	400	8
4	250	25	450	10
5	300	30	500	12
6	350	10	550	14
7	400	15	600	16
8	450	20	650	18
9	500	25	700	20
10	550	30	750	10
11	600	10	800	12
12	650	15	300	14
13	700	20	350	16
14	50	25	400	18
15	100	30	450	20
...	...	...	...	...
992	200	15	550	6
993	250	20	600	8
994	300	25	650	10
995	350	30	700	12
996	400	10	750	14
997	450	15	800	16
998	500	20	300	18
999	550	25	350	20
1000	600	30	400	10

From:

N learning examples: $\{(Input_k, Output_k)\}_{k=1}^{N}$

$Output_k = \mathcal{F}_w(Input_k)$ or $n = \mathcal{F}_w(P, V, T)$



$$n = \frac{P.V}{R.T}$$

$$n = tanh\left(tanh\left(P \times w_{P_c_1} + V \times w_{V_c_1} + T \times w_{T_c_1} + w_{b_c_1} \right) \times w_{c_1_n} + tanh\left(P \times w_{P_c_2} + V \times w_{V_c_2} + T \times w_{T_c_2} + w_{b_c_2} \right) \times w_{c_2_n} + w_{b_n} \right)$$

UNIVERSITÉ SORBONNE PARIS NORD | Institut GALILÉE Université Sorbonne Paris Nord

Reality is more complex!

Recommendations at Amazon

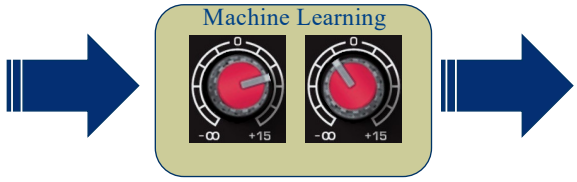
A personalized space that is highlighted on the homepage, and it is surely not by chance.

Recommendations at Amazon

To make recommendations, Amazon uses all the information it has:

Input

- The products I bought;
- The products that I said I already have;
- The products I consulted
- Links between products, extrapolated from shopping lists, created by all customers.
- ...



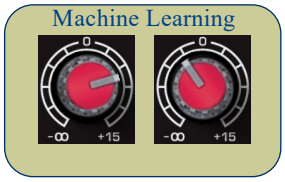
Output

Will you like, click, comment, interact, ...?

Detection and investigation of credit card fraud on the Internet

Input

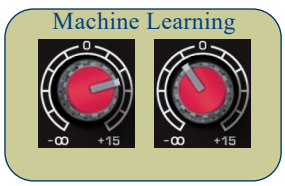
Type	Nb
Variables GIE	37
Agrégats Carte	304
Agrégats Marchand	370
Variables Sociales Carte	140
Variables Sociales Marchand	41
Score Carte et Agrégats Score Carte	17
Score Marchand et Agrégats Score Marchand	17
Score OCSVM (linéaire)	19
Score OCSVM (gaussien)	19
Total	964



Output

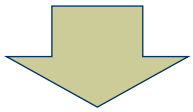
Fraud?

Image recognition



- Flower?
- Flamingo?
- Lamp?
- Robot?
- ...?

For real problems in large size, the regression line is not enough!

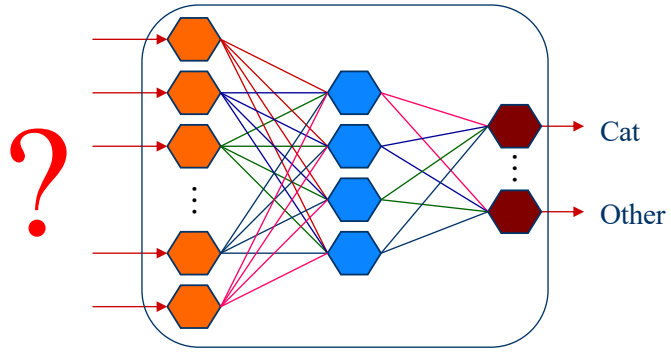


More complex and richer models: artificial neural networks

Large size vs. complexity

Image 400x400

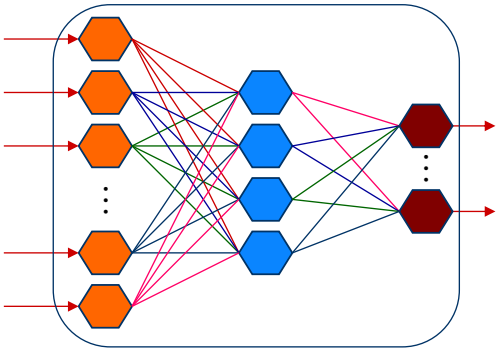
160 000 numbers as input



Extract features

Image 1000x1000

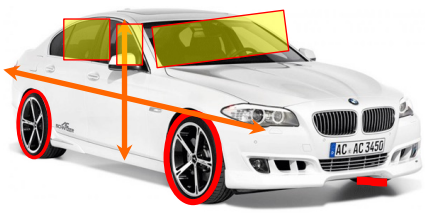
?



- Motorbike?
- Car ?
- Bus ?
- Train ?
- … ?

10^6 numbers
as input

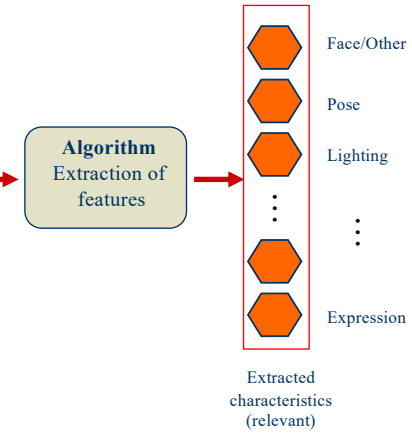
Extract features



Discovering the hidden structure of large dimension data

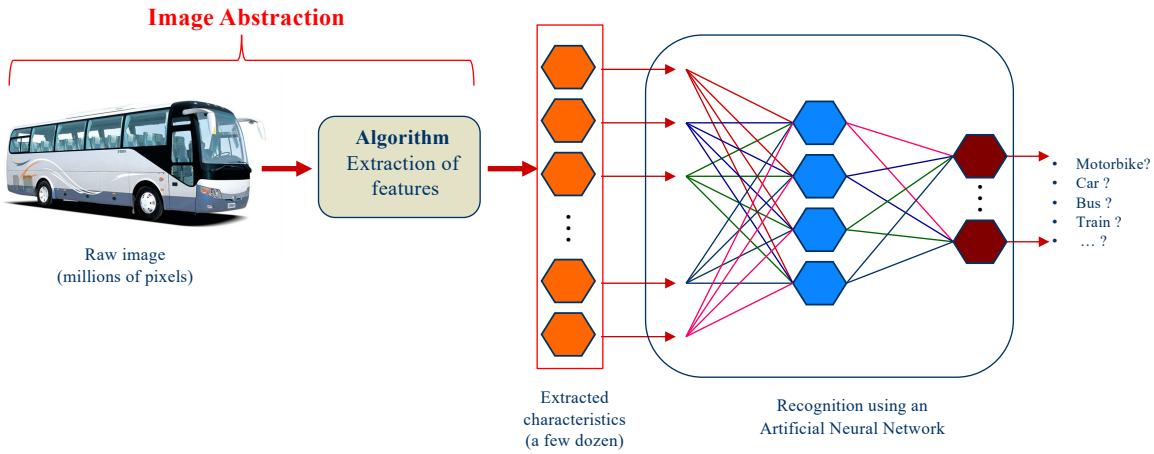
Example: images of a person's face
1000x1000 pixels = 1,000,000 dimensions

- **But the face has 3 Cartesian coordinates and 3 Euler angles.**
- **The human face has about 50 muscles**
- **The hidden dimension < 56 dimensions**

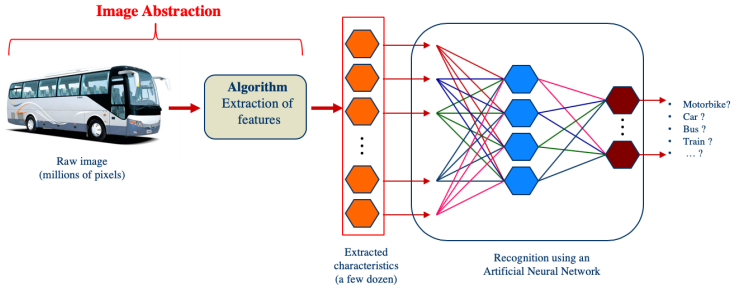


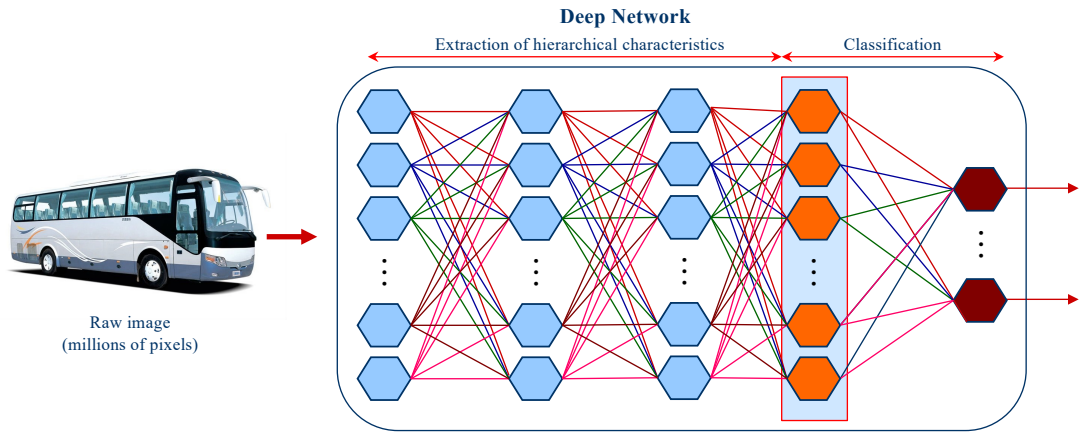
The relevant representation of a face image must contain less than 10^6 values!

Extract features

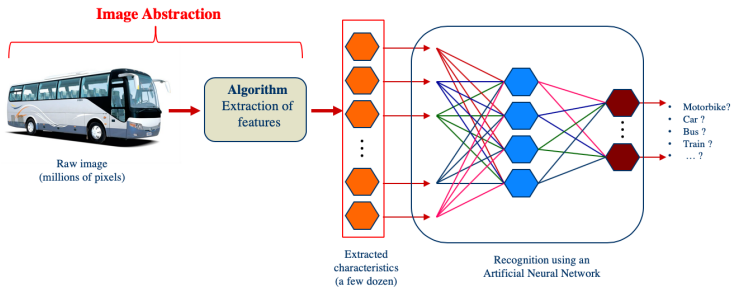


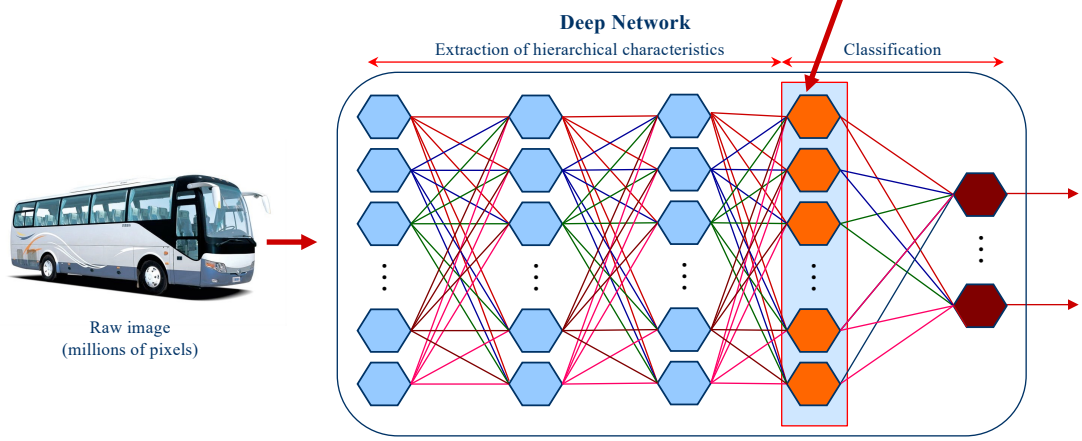
Extract features



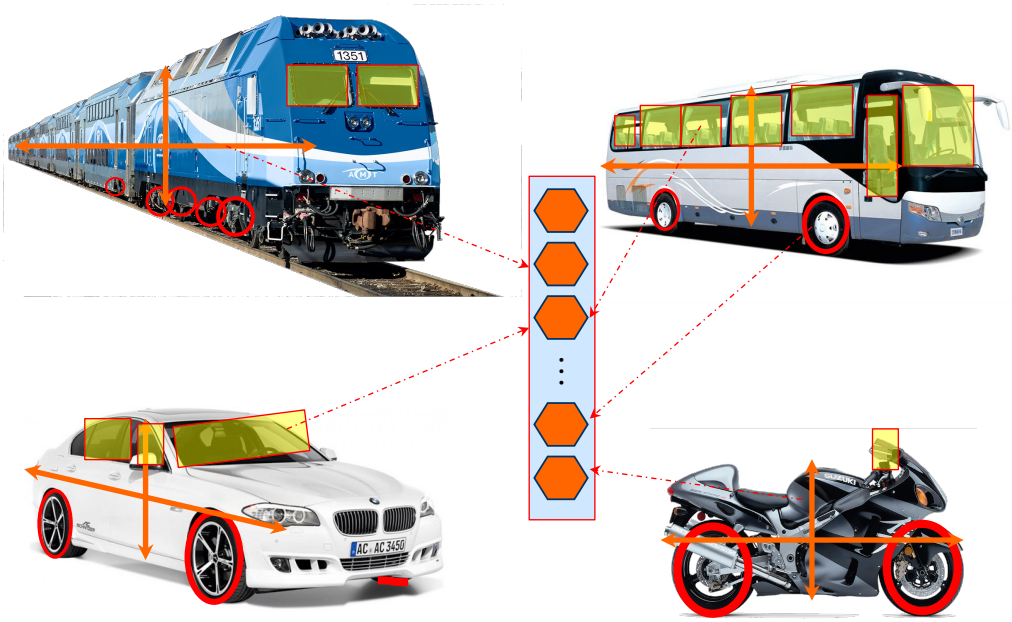


Learning the characteristics





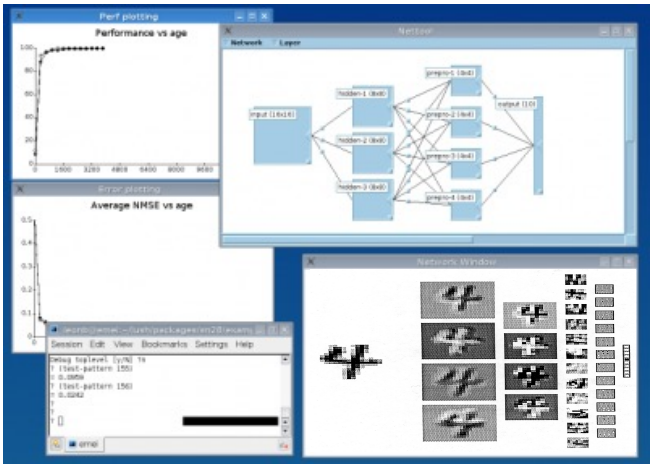
Learning the characteristics



How to learn features with such networks?

Yann LE CUN

Léon BOTTOU



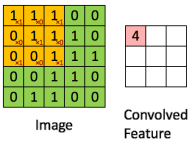

```
% sn28

SN2.8 (C) Copyright Neuristique s.a. 91 92 93
Contributors: [ Y. Bennani, L. Bottou, J. Bourrely,
                X. Driancourt, Y. Le Cun, S. Suddarth ]
Version: 'sn28'
Licensed to: NSA
... loading system file "/usr/local/sn2.8/lib/sysenv.sn"
... loading system file "/usr/local/sn2.8/lib/stdenv.sn"
 [graphenv.sn]
 [netenv.sn]
 [knnenv.sn]
 [connect.sn]
```

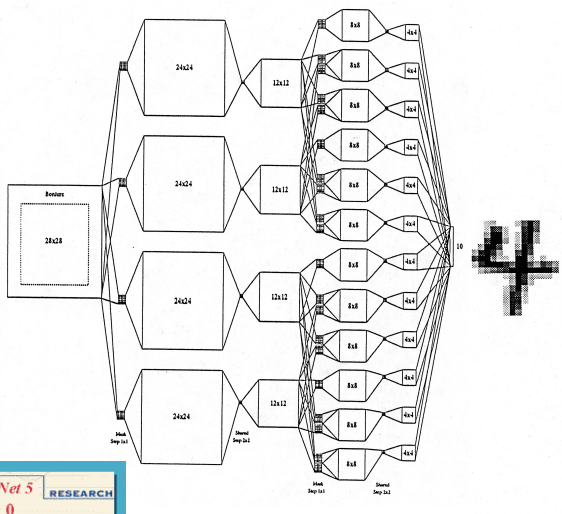
Neuristique s.a.

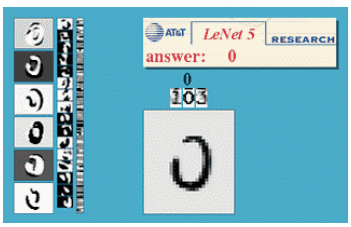
Neuristique was founded in 1988 by a dozen friends with big dreams. The mission statement was very long sentence that mentions the application of artificial neural networks, the development of artificial brains, and the exploration of space.

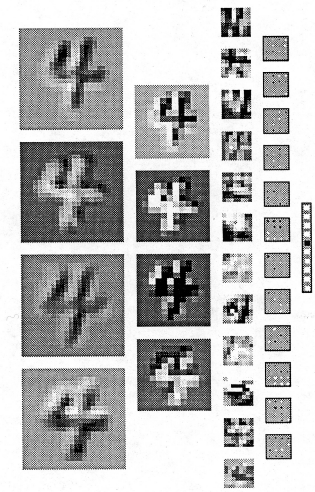
How to learn features with such networks?

How to learn features with such networks?







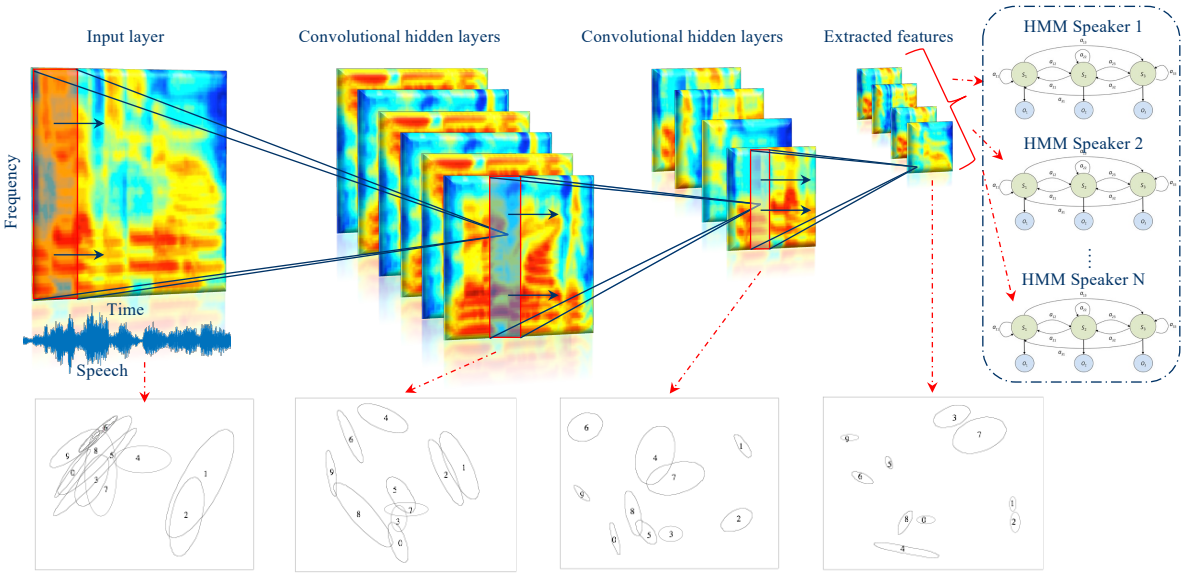


Le Cun Y., Boser B., Denker J.S., Henderson D., Howard R.E., Hubbard W., Jackel L.D. (1989) : «Back-propagation applied to handwritten zip code recognition» Neural Computation, Vol. 1, pp. 541-551.

How to learn features with such networks?

Younès BENNANI

Patrick GALLINARI



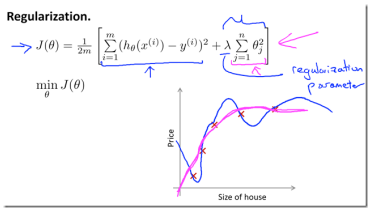
BENNANI Y. & GALLINARI P. (1991), "*On The Use Of TDNN-Extracted Features Information In Talker Identification*", IEEE International Conference on Acoustics, Speech, and Signal Processing, ICASSP '91, May 14-17, pp. 385-388, Toronto, CANADA

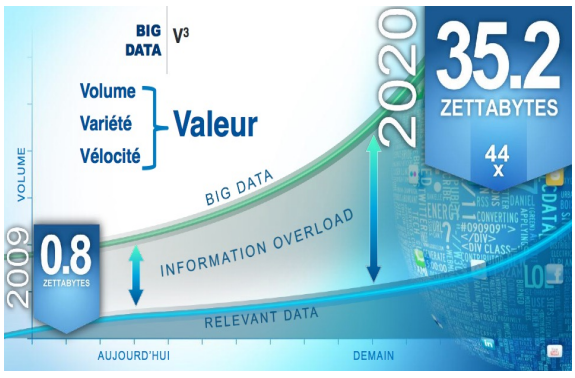
Turing Award 2018 honors a different kind of AI

IBM Google Meta

Nobel Prize in Physics & Chemistry 2024

Why this strong comeback?





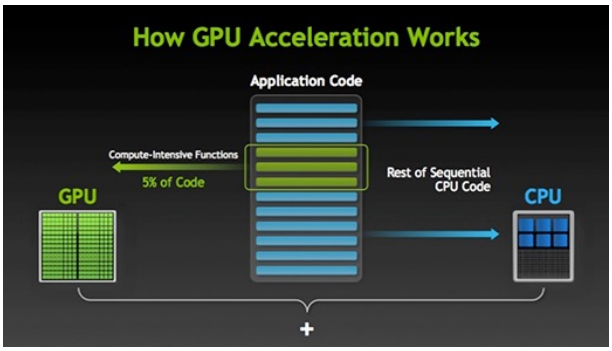
Big Data: data in all directions ...
An exponential explosion in the quantity of data

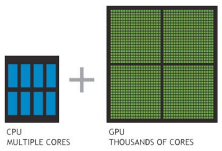
NVIDIA DGX-1
Deep Learning System

Graphics Processing Units

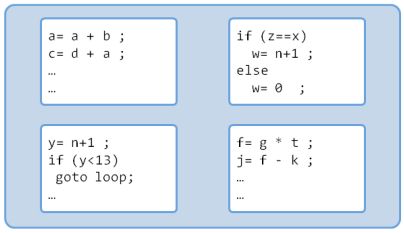
GPU computing: using the graphics processing unit (GPU) in parallel with the CPU to accelerate computations

How does the GPU work?

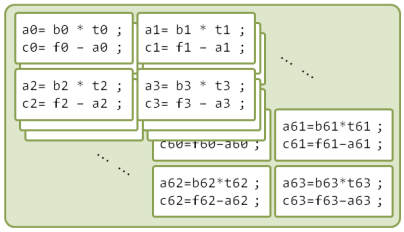




CPU quadcore exécutant simultanément 4 **programmes différents** sur des **données différentes**



GPU exécutant simultanément 64 **programmes identiques** sur des **données différentes**, sur 64 *cores*



The graphics card becomes a "coprocessor" that supports the main processor of a system.

CPU : Central processing unit

GPU : Graphics processing unit

Learning in Deep Networks

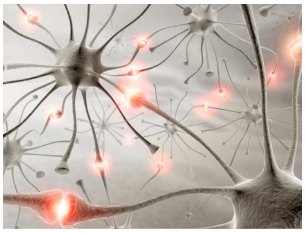
- Constructing a feature space
  - Note that this is what we do with kernels in SVM, or hidden layers in MLP, etc, but now we will build the representation space using deep architectures.
  - Unsupervised learning between layers can decompose the problem into distributed sub-problems (with higher levels of abstraction) to be further decomposed at successive layers.

The challenge of learning in deep networks

- Difficulties in supervised learning of deep networks
  - The first layers of MLP are not well learned
    - Gradient Diffusion - error is reduced as it propagates to the previous layers: the gradient propagates "badly" from the output to the input.
    - Leads to very slow learning.
    - The deeper the network, the greater the degree of non-linearity of the network, which would increase the chances of finding these obstacles to optimisation.
    - The lower layers remain with not very useful transformations of the input.
    - Need a way to help the first layers to do an efficient job.
  - Often not enough labelled data available
    - Can we use unsupervised/semi-supervised approaches to take advantage of untagged data?
  - Deep networks tend to have more local minima problems than simple networks during supervised learning

The original idea (≈1940)

• Living organisms, even primitive ones (eg insects), carry out **complex tasks** of **information processing**:
  • Orientation
  • Communication
  • Social behavior
  • ...



• The **processing power** of their **nervous system** comes from:
  • Massive interconnection (10^{14} connections in humans)
  • Large number of single treatment units (10^{11} in humans): neurons

• The **initial motivation** was to make neuromimicry
  • however, the 1940s vision was rather simplistic;
  • the **biological reality** has been **more complex** since.

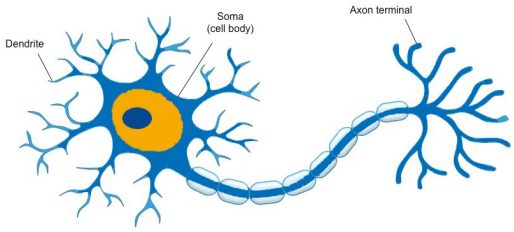
• On the other hand, this **idea** has proved very **fruitful in mathematics** and **engineering**.

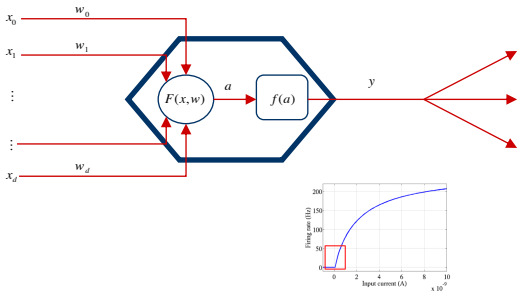
The artificial neuron

1943, Mc Culloch & Pitts

A model and not a copy of the biological neuron
= an elementary processor characterized by:

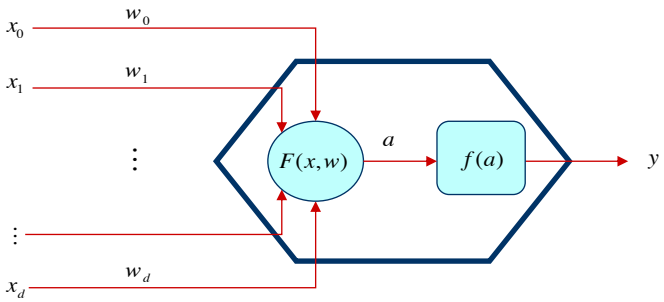
- **input signals** $x = x_0, x_1, \ldots, x_d$
- **weight of connections** $w = w_0, w_1, \ldots, w_d$
- **activation function** $F(x,w)$
- **internal activation state** $a = F(x,w)$
- **transition function** $f(a)$
- **output state** $y = f(a)$

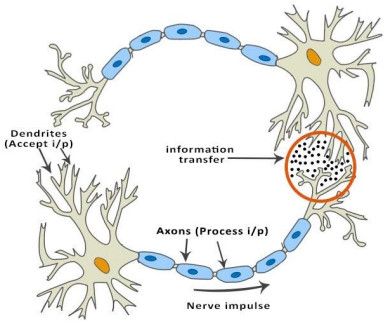




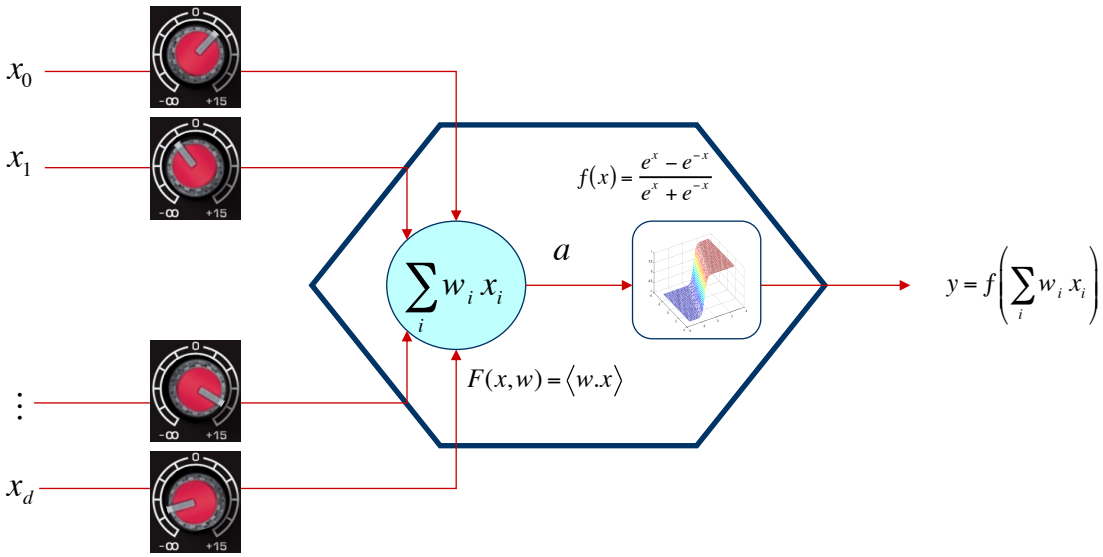
The artificial neuron

Definition: *A formal (artificial) neuron is a processing unit that receives input data, in the form of a vector, and produces a real output. This output is a function of the inputs and the weights of the connections.*

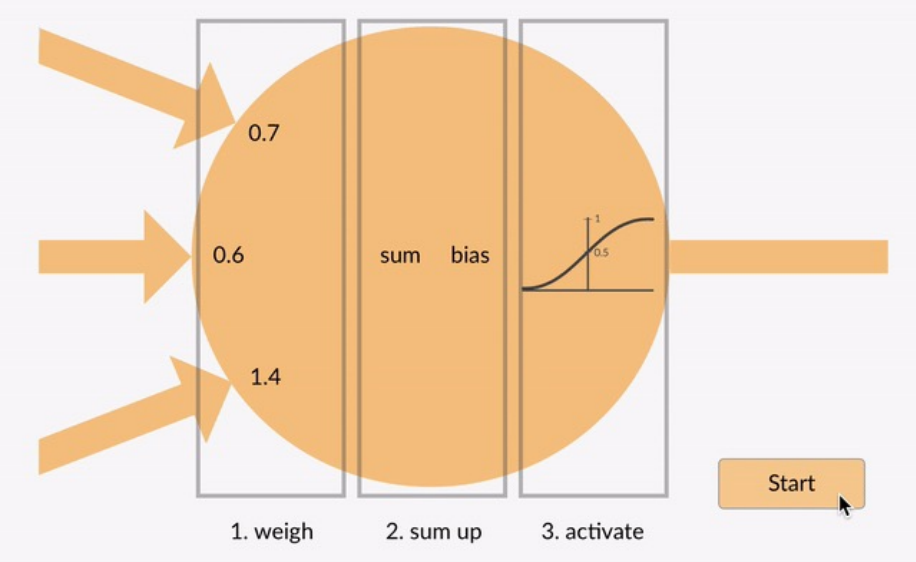




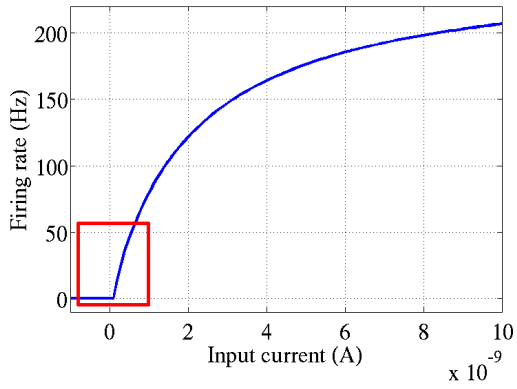
The artificial neuron

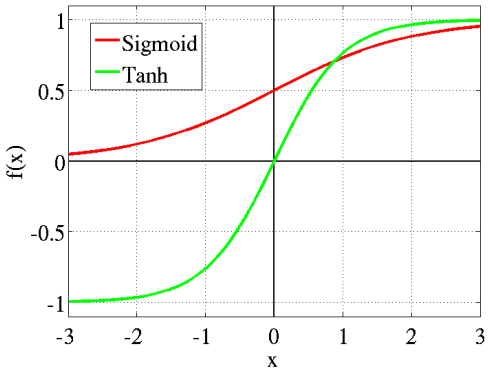


The artificial neuron

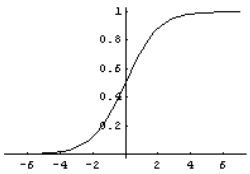


Activation functions

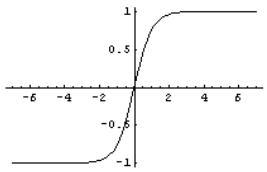




Left: **Common neural activation function motivated by biological data.** *Right:* **Commonly used activation functions** in neural networks literature: logistic sigmoid and hyperbolic tangent (*tanh*).

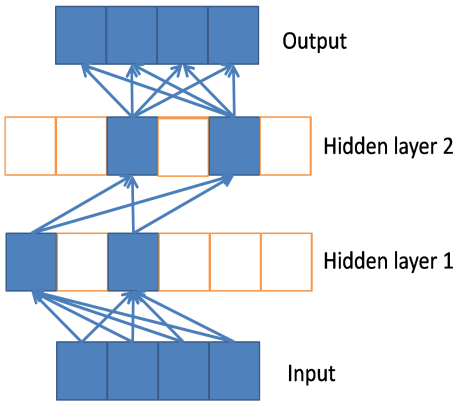


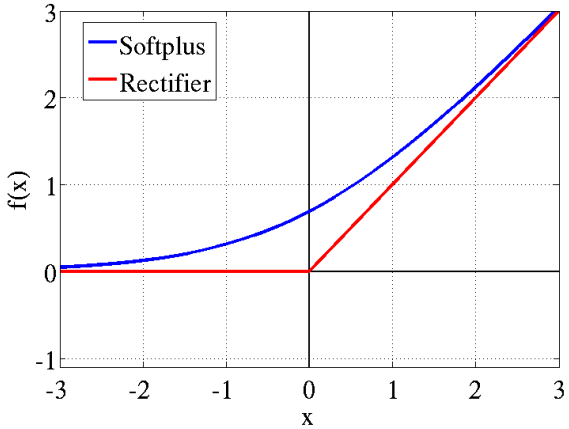
$$f(x) = \frac{1}{1+e^{-x}} \quad \rightarrow f'(x) = f(x)\big(1-f(x)\big)$$



$$f(x) = \tanh(x) = \frac{e^{x}-e^{-x}}{e^{x}+e^{-x}} \qquad f(x) = \frac{e^{x}-1}{e^{x}+1}$$

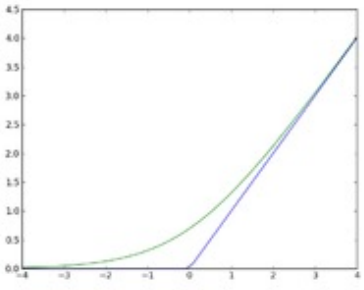
Activation functions

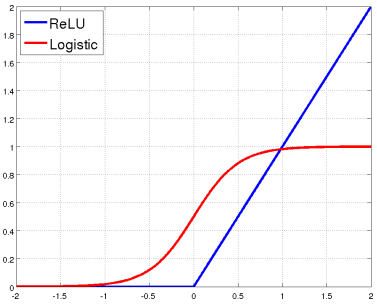




I *Left:* **Sparse propagation of activations and gradients in a network of rectifier units.** The input selects a subset of active neurons and computation is linear in this subset. *Right:* **Rectifier and softplus activation functions.** The second one is a smooth version of the first.

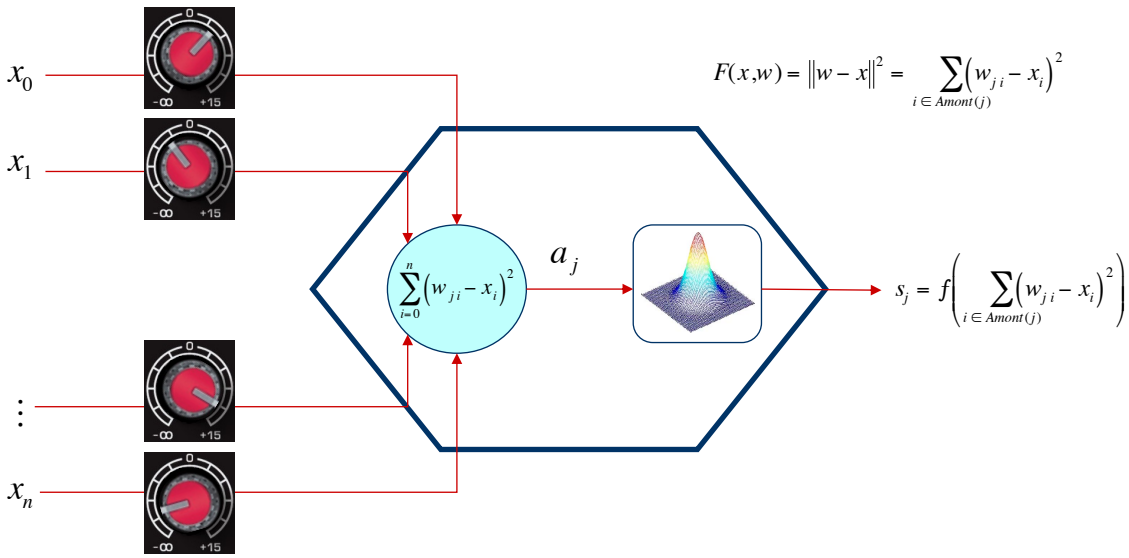
Rectified Linear Units





- More efficient gradient propagation, derivative is 0 or constant, just fold into learning rate
- More efficient computation: Only comparison, addition and multiplication.
  - Leaky ReLU: $f(x) = x$ if > 0 else ax where $0 \leq a <= 1$, so that derivate is not 0 and can do some learning for that case.
  - Lots of other variations
- Sparse activation: For example, in a randomly initialized networks, only about 50% of hidden units are activated (having a non-zero output)

The Neuron Distance



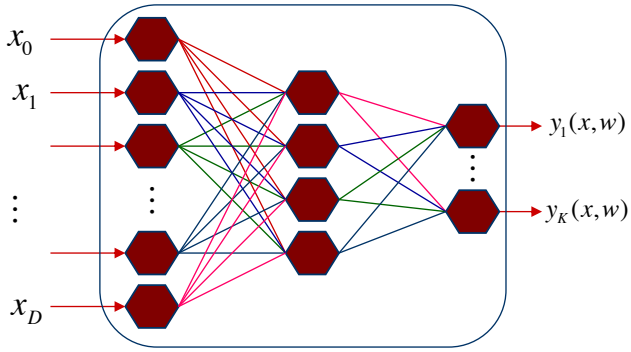
$$\|w-x\|^2 = \|w\|^2 - 2\langle w,x\rangle + \|x\|^2$$

Artificial Neural Network

Definition: *A neural network is a valued oriented graph consisting of a set of units (or automata), performing elementary computations, structured in successive layers capable of exchanging information by means of connections that connect them.*

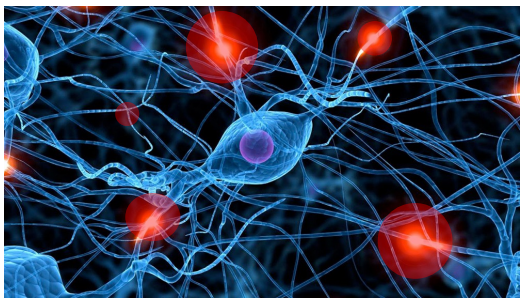
- Massively parallel architecture.
- System based on the cooperation of several simple units (formal neurons).

- A network is characterized by:
  - architecture
  - functions of its elements

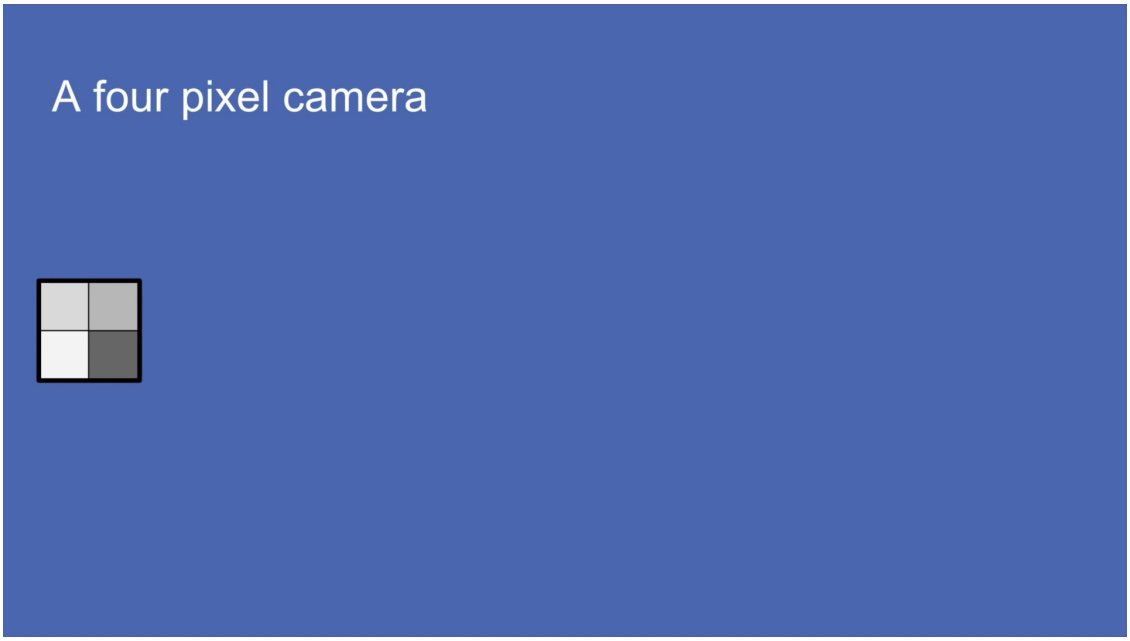


Artificial Neural Network

Adaptation network parameters≡ Synaptic efficiency



How Artificial Neural Networks work?



Adaline : Adaptive Linear Element

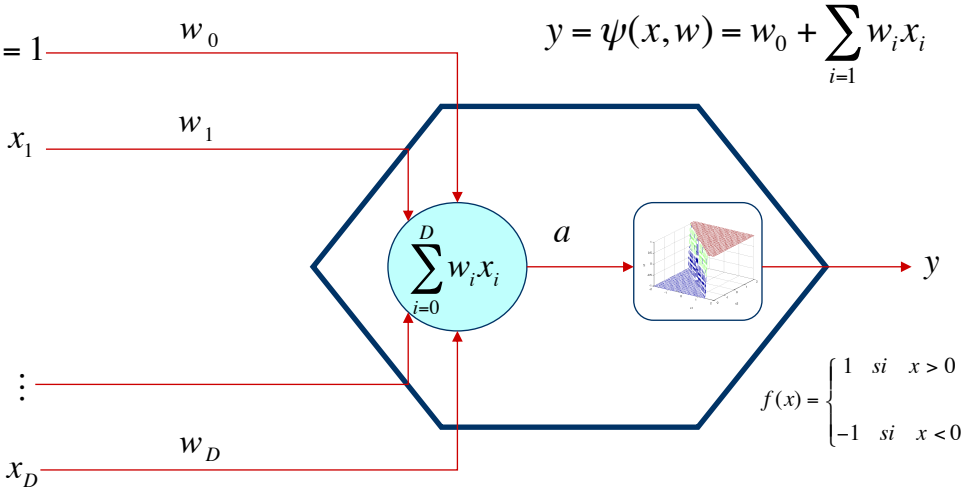
Stanford, 1960, Bernard Widrow*

It is an adaptive linear element:

$$y = \sum_{i=0}^{D} w_i x_i$$

The unit x_0, whose activation set to 1, is called bias unit. It allows to introduce the terms of bias in the network.

$$y = \psi(x, w) = w_0 + \sum_{i=1}^{D} w_i x_i$$



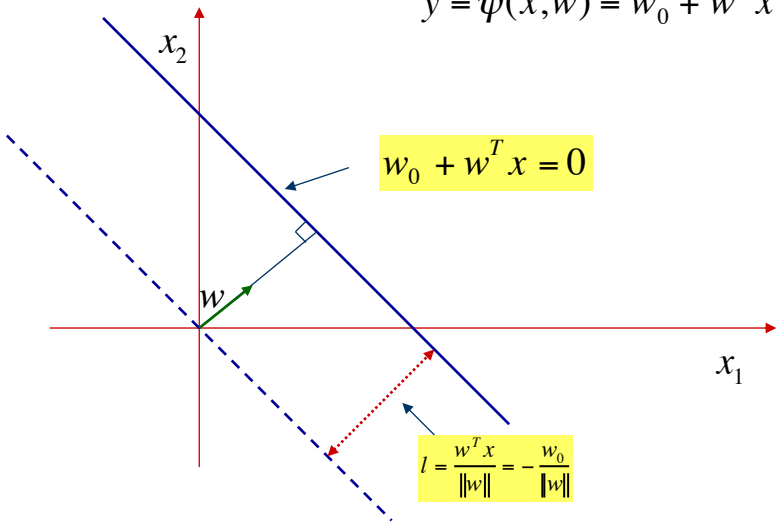
* Widrow B., Hoff M.E. (1960) : « Adaptive switching circuits », IRE WESCON Conv. Record, part 4, pp. 96-104.

Adaline : Adaptive Linear Element

Stanford, 1960, Bernard Widrow*

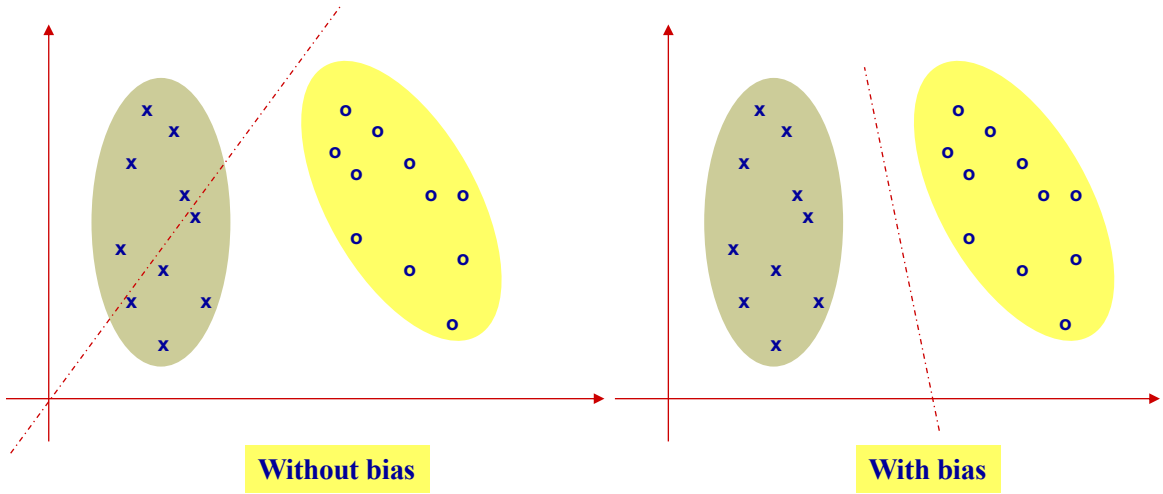
$$y = \psi(x, w) = w_0 + \sum_{i=1}^{D} w_i x_i$$

$$y = \psi(x, w) = w_0 + w^T x$$



Adaline : Adaptive Linear Element

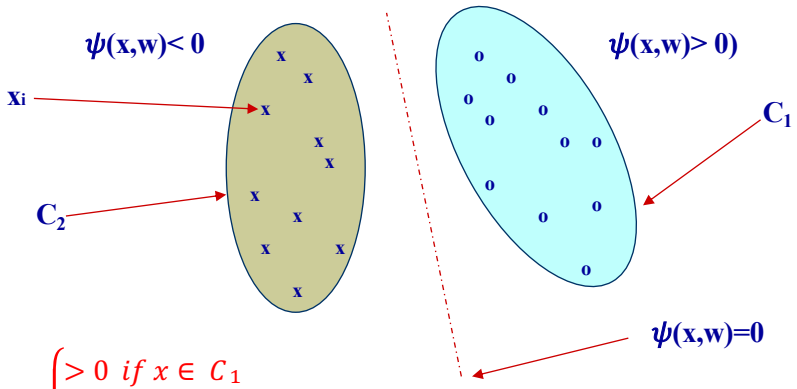
Stanford, 1960, Bernard Widrow*



Adaline : Adaptive Linear Element

Stanford, 1960, Bernard Widrow*

It can be used in discrimination (classification) between 2 classes:



$$\psi(x,w) = w_0 + w^T x \begin{cases} > 0 \ \ if \ x \in C_1 \\ < 0 \ \ if \ x \in C_2 \end{cases}$$

Adaline : Adaptive Linear Element

Stanford, 1960, Bernard Widrow*

If we call $z^{(n)} = \left(x^{(n)}, d^{(n)}\right)$ the example taken into account at iteration n,
We define the square of the instantaneous error by:

$$\tilde{R}^{(n)}_{Adaline}(w) = \left(d^{(n)} - w\, x^{(n)}\right)^2$$

Desired output — Computed output: y

The mean squared error or (MSE) is defined as the mean observed squares of instantaneous errors on all of all examples:

$$\tilde{R}_{Adaline}(w) = \frac{1}{N}\sum_{n=1}^{N} \tilde{R}^{(n)}_{Adaline}(w)$$

There are several learning algorithms.

Adaline : Adaptive Linear Element

Stanford, 1960, Bernard Widrow*

Gradient descent techniques (the biggest slope):
suppose that at the moment t, the weights of the Adaline are $w(t)$
and show the example $(x^{(n)}, d^{(n)})$, then the weights will be changed by:

$$w(t+1) = w(t) - \varepsilon(t)\nabla_w \tilde{R}^{(n)}_{Adaline}(w)$$

The step of the gradient

The instant gradient

$$\nabla_w \tilde{R}^{(n)}_{Adaline}(w) = \frac{\partial \tilde{R}^{(n)}_{Adaline}(w)}{\partial w} = -2\left(d^{(n)} - w\, x^{(n)}\right)x^{(n)}$$

This rule is called stochastic gradient rule
or Widrow-Hoff rule
or delta Widrow-Hoff rule
or μ-LMS (Least Mean Square) rule

Gradient descent techniques

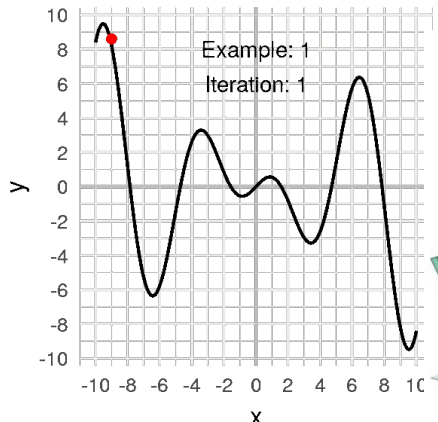
Gradient descent algorithm

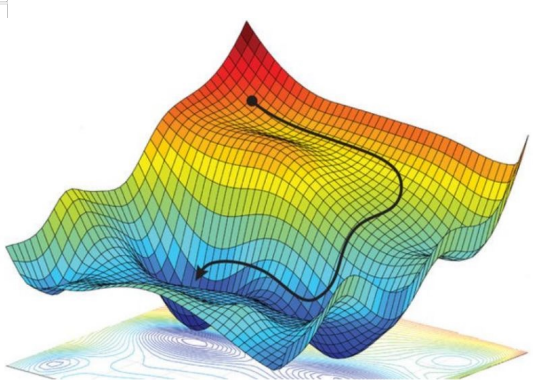
repeat until convergence {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

(simultaneously update $j = 0$ and $j = 1$)

}





Gradient descent techniques

Gradient descent algorithm

repeat until convergence {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

(simultaneously update $j = 0$ and $j = 1$)

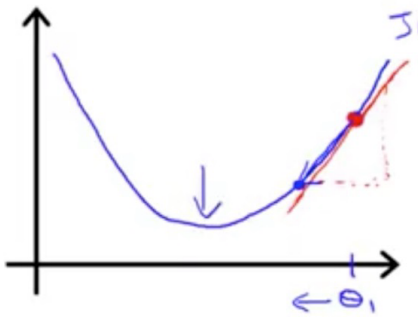
}

learning rate

derivative

$\min_{\theta_1} J(\theta_1)$ $\quad \theta_1 \in \mathbb{R}$.

Gradient descent techniques



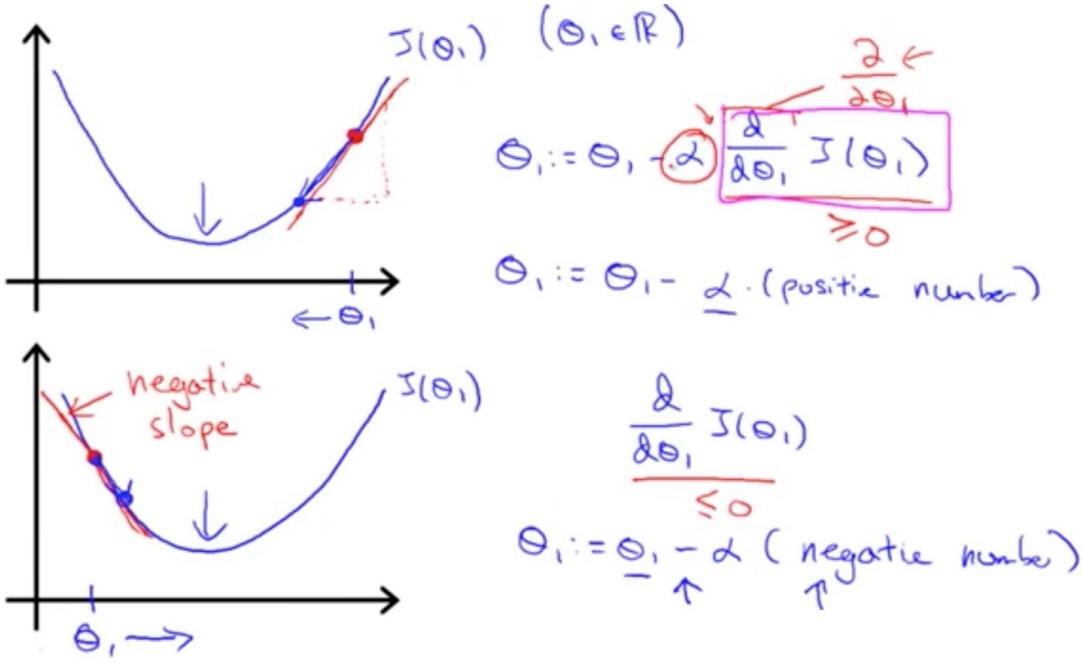
$J(\theta_1)$ $(\theta_1 \in \mathbb{R})$

$$\theta_1 := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_1)$$

$\frac{\partial}{\partial \theta_1}$: ≥ 0

$\theta_1 := \theta_1 - \alpha \cdot$ (positive number)

Gradient descent techniques



Adaline : Adaptive Linear Element

Stanford, 1960, Bernard Widrow*

Learning algorithm ADALINE

1- Initialize randomly W_0

2- Randomly choose a couple of data $\left(x^{(n)}, d^{(n)}\right)$

3- Determine the value of the error

$$e^{(n)}(t) = \left(d^{(n)} - w\, x^{(n)}\right)$$

4- Calculate a gradient approximation

$$\nabla_w \tilde{R}^{(n)}_{Adaline}(w) = -2e^{(n)}(t)\, x^{(n)}$$

5- Adapt the weights $w(t)$

$$w(t+1) = w(t) - \varepsilon(t)\nabla_w \tilde{R}^{(n)}_{Adaline}(w)$$

Where $\varepsilon(t)$ is de step of the gradient.

6- Repeat from 2 to 4 until an acceptable error value is obtained.

Adaline : Example

Data: Truth Table of a Boolean Function of 2 Variables

x_1	x_2	d
1	1	1
1	-1	1
-1	1	1
-1	-1	-1

Function : $\psi(x,w)= (x_1 \text{ or } x_2)$

$$D_4 = \left\{ \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}, 1 \right); \left(\begin{pmatrix} 1 \\ -1 \end{pmatrix}, 1 \right); \left(\begin{pmatrix} -1 \\ 1 \end{pmatrix}, 1 \right); \left(\begin{pmatrix} -1 \\ -1 \end{pmatrix}, -1 \right) \right\}$$

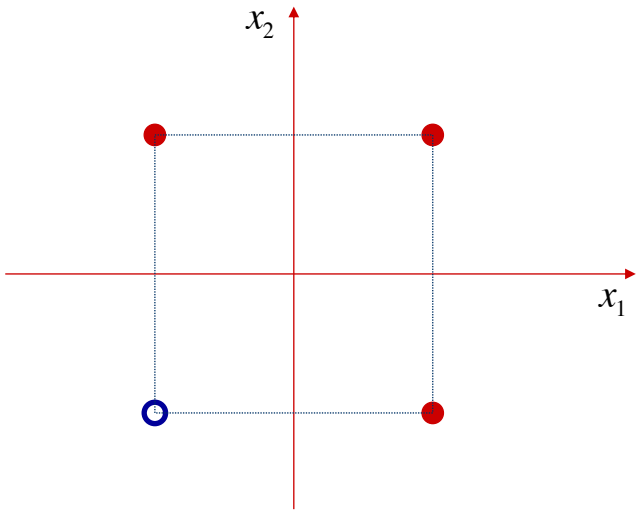
Problem:

Find an Adaline able to learn the truth table of a Boolean function of 2 variables.

Adaline : Example

Function : $\psi(x,w)= (x_1 \text{ or } x_2)$

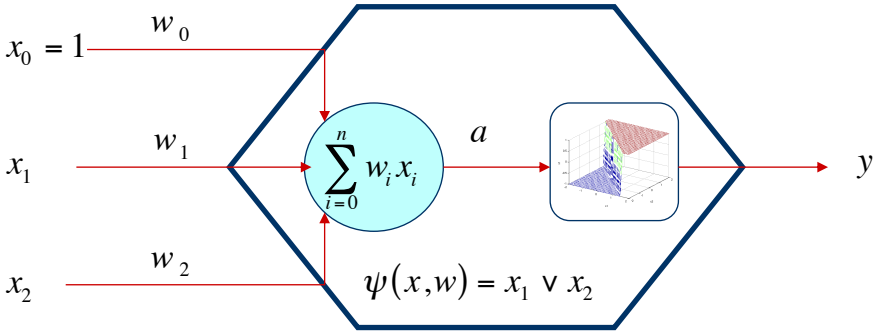
x_1	x_2	d
1	1	1
1	-1	1
-1	1	1
-1	-1	-1



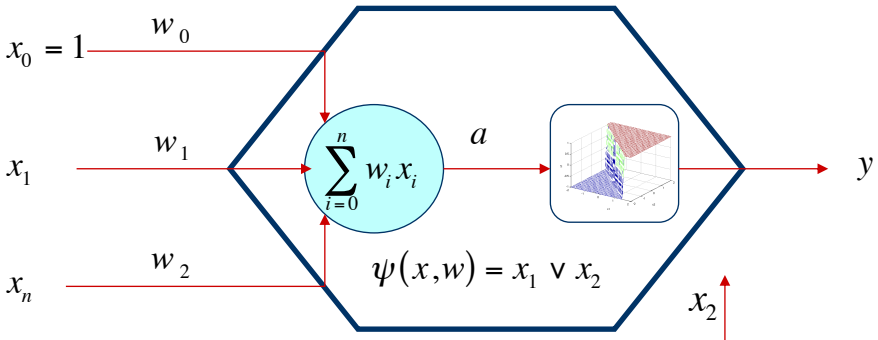
Adaline : Example

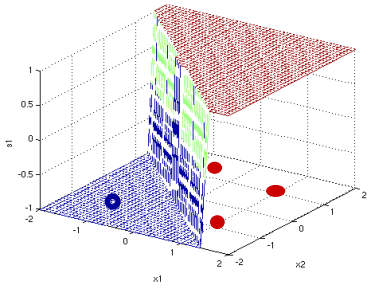
$$D_N = \left\{\left(x^{(1)}, d^{(1)}\right), \left(x^{(2)}, d^{(2)}\right), ..., \left(x^{(N)}, d^{(N)}\right)\right\}$$

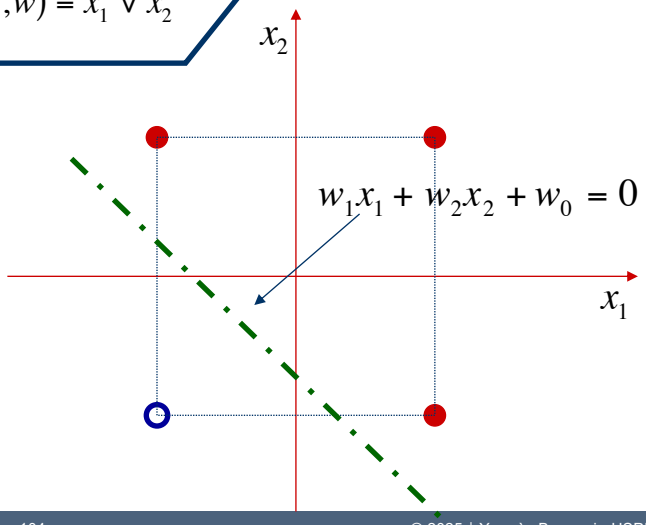
x_0	x_1	x_2	d
1	1	1	1
1	1	-1	1
1	-1	1	1
1	-1	-1	-1



Adaline : Example



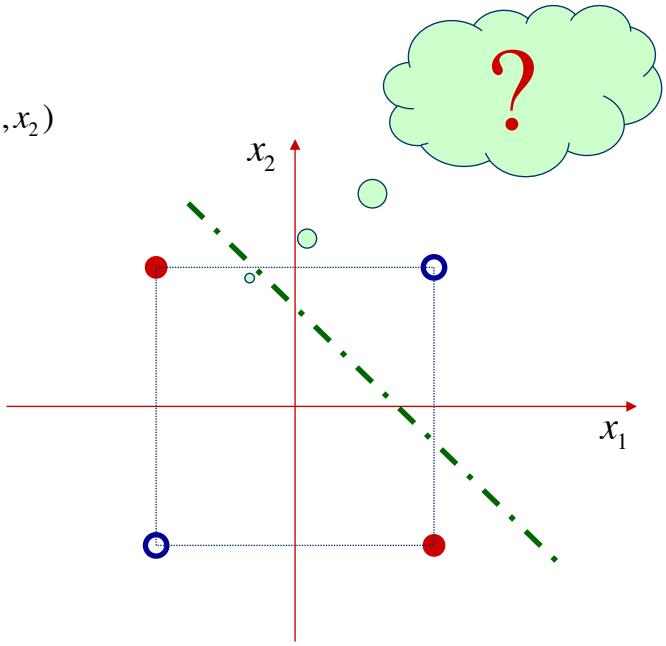




Adaline : limitations

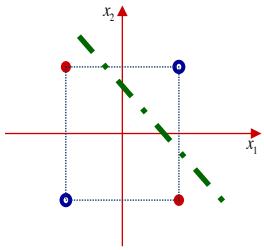
$$\psi(x,w) = x_1 \oplus x_2 = XOR(x_1,x_2)$$

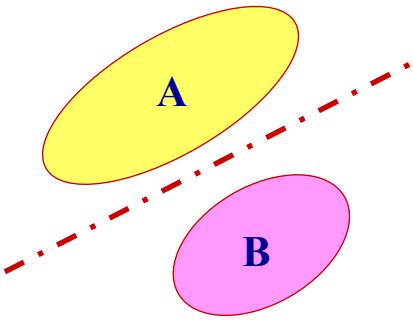
	x_0	x_1	x_2	d	
x^1	1	1	1	-1	○
x^2	1	1	-1	1	●
x^3	1	-1	1	1	●
x^4	1	-1	-1	-1	○



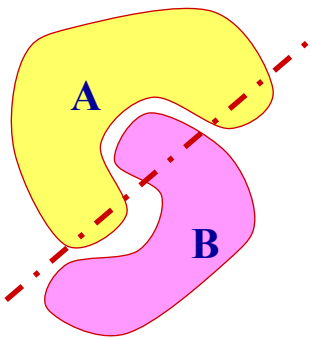
Linear separability

« Two classes of objects, described in a space of dimension n, are said to be "linearly separable" if they are on both sides of a hyperplane in the representation space. »





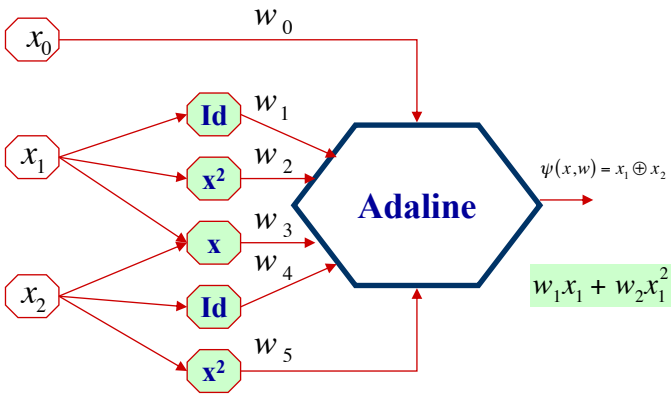
Linearly separable



Non-Linearly separable

Madaline : Multi-Adaptive Linear Element

Adaline with polynomial preprocessing

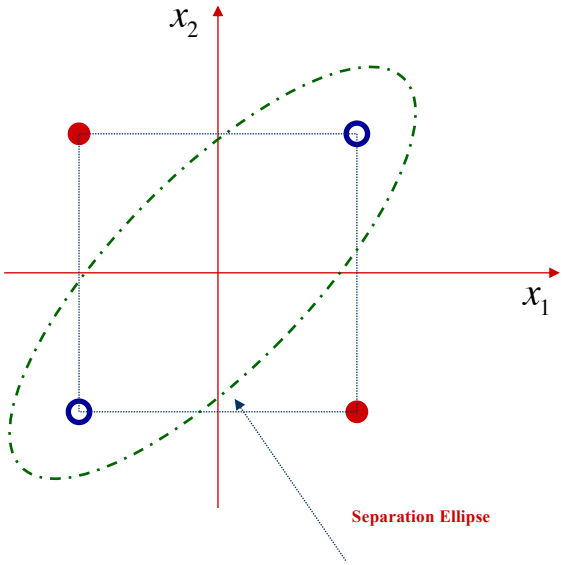


$$w_1 x_1 + w_2 x_1^2 + w_3 x_1 x_2 + w_4 x_2 + w_5 x_2^2 + w_0 = 0$$

UNIVERSITÉ SORBONNE PARIS NORD | Institut GALILÉE Université Sorbonne Paris Nord

Madaline : Multi-Adaptive Linear Element

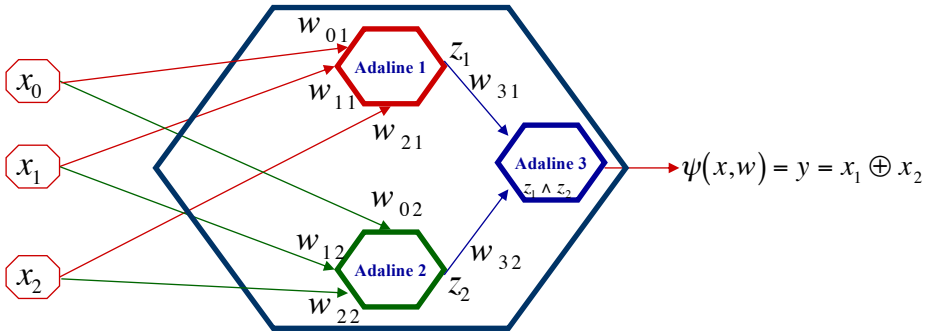
Adaline with polynomial preprocessing



$$w_1 x_1 + w_2 x_1^2 + w_3 x_1 x_2 + w_4 x_2 + w_5 x_2^2 + w_0 = 0$$

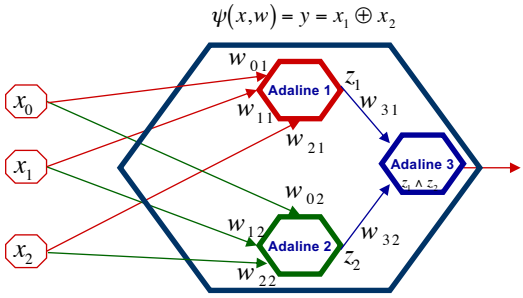
Madaline : Multi-Adaptive Linear Element

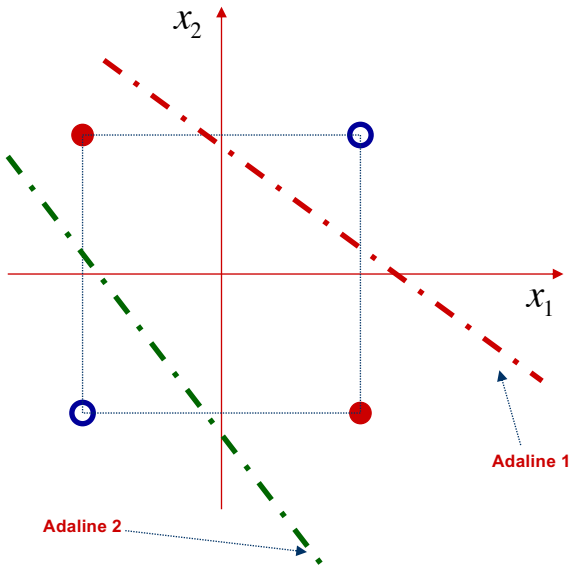
Madaline = a set of parallel Adalines



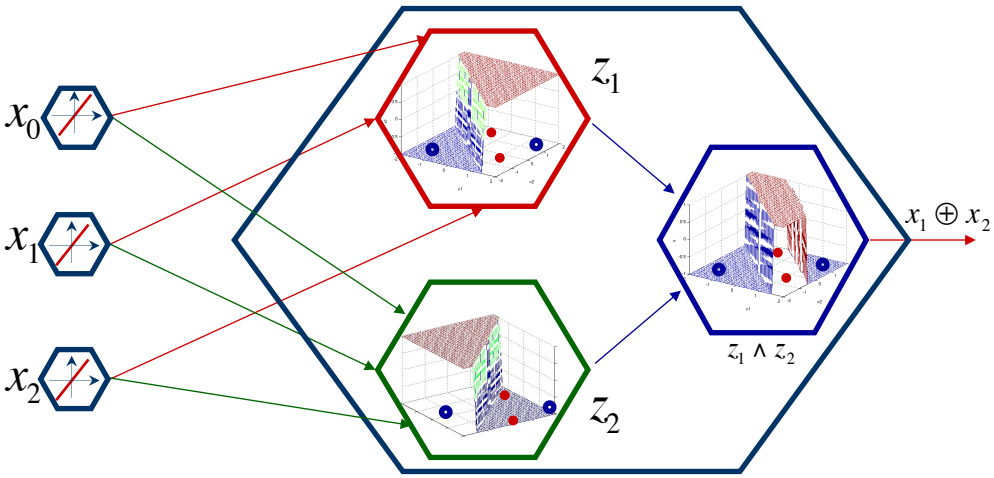
	x_0	x_1	x_2	Adaline 1: d	Adaline 2: d	Adaline 3: d
x^1	1	1	1	-1	1	-1
x^2	1	1	-1	1	1	1
x^3	1	-1	1	1	1	1
x^4	1	-1	-1	1	-1	-1

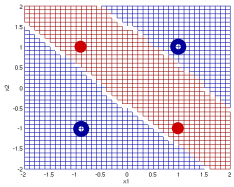
Madaline : Multi-Adaptive Linear Element





Madaline : Multi-Adaptive Linear Element





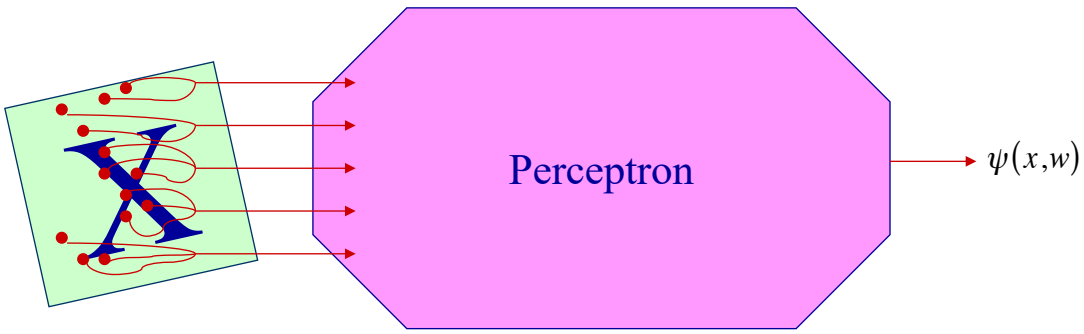
Perceptron

Rosenblatt F., 1957, 1962*

1957, Frank Rosenblatt

The perceptron does not designate a single model but groups together an important family of algorithms.

The perceptron = adaptive machine used to solve classification problems (discrimination).



* Rosenblatt F. (1957) : « The perceptron: a perceiving and recognizing automaton », Reports 85-460-1, Cornell Aeronautical Lab., Ithaca, N.Y.
* Rosenblatt F. (1962) : « Principles of Neurodynamics: perceptrons and theory of brain mechanisms », Spartan Books, Washington.

Perceptron

Rosenblatt F., 1957, 1962

The Retina R

Receives the information from the outside

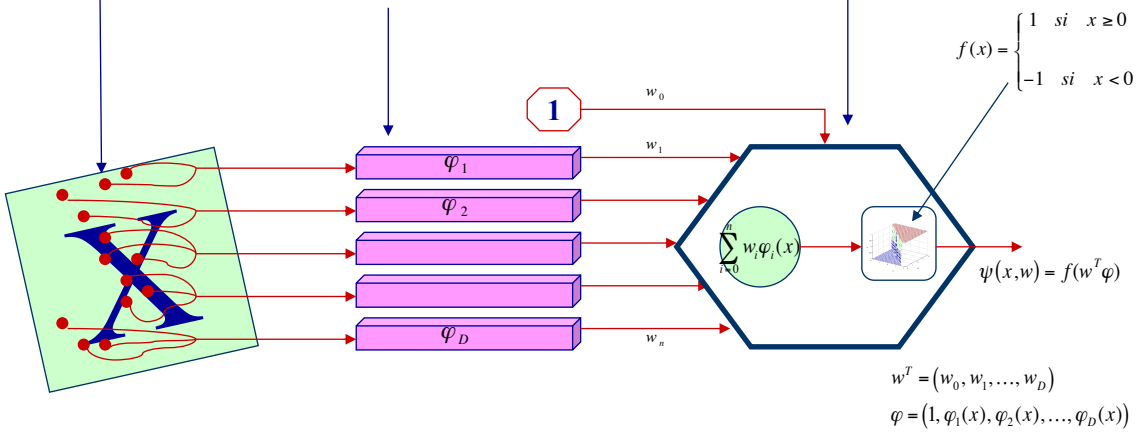
The association cells

each cell has a function transition defined on the retina:

$$\varphi_i(x): \quad R \rightarrow \Re$$

Decision cell

$$\psi(x,w) = f\left(w_0 + \sum_{i=1}^{D} w_i\, \varphi_i(x)\right)$$

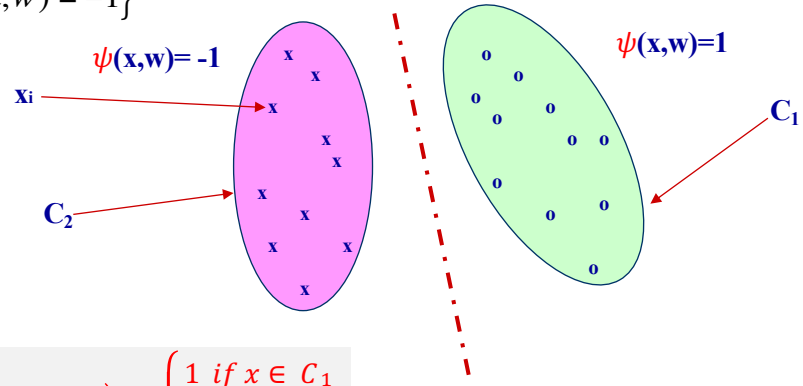


Perceptron

2-class case

A Perceptron can be seen as a 2-class classifier:

$$C_1 = \{x \in R : \psi(x,w) = 1\}$$
$$C_2 = \{x \in R : \psi(x,w) = -1\}$$



$$\psi(x,w) = f\left(w_0 + \sum_{i=1}^{D} w_i\varphi_i(x)\right) = \begin{cases} 1 \ if \ x \in C_1 \\ -1 \ if \ x \in C_2 \end{cases}$$

Perceptron

2-class case

If we call $(x^{(n)}, d^{(n)})$ the example taken into account at iteration n, we define the square of the instantaneous error by:

$$\begin{cases} d^{(n)} = 1 \ for \ x^{(n)} \in C_1 \\ d^{(n)} = -1 \ for \ x^{(n)} \in C_2 \end{cases} \Rightarrow \begin{cases} w^T\varphi^{(n)} > 0 \ for \ x^{(n)} \in C_1 \\ w^T\varphi^{(n)} < 0 \ for \ x^{(n)} \in C_2 \end{cases} \Rightarrow \forall x^{(n)}, w^T\left(\varphi^{(n)} d^{(n)}\right) > 0$$

$$\Downarrow$$

$$\tilde{R}^{(n)}(w) = -w^T\left(\varphi^{(n)} d^{(n)}\right)$$

The mean square error (MSE) is:

$$\tilde{R}(w) = \frac{1}{N}\sum_{n=1}^{N} \tilde{R}^{(n)}(w) = -\sum_{\{k:x^{(n)}\ misclassified\}} w^T\left(\varphi^{(n)} d^{(n)}\right)$$

There are several learning algorithms.

Perceptron

2-class case

Gradient descent techniques (the biggest slope):
suppose that at the moment t , the weights of the Perceptron are $w(t)$ and show the example $(x^{(n)}, d^{(n)})$, then the weights will be changed by:

$$w(t+1) = w(t) - \varepsilon(t)\nabla_w \tilde{R}(w)$$

Step of the gradient

Instant gradient

$$\nabla_w \tilde{R}(w) = \frac{\partial \tilde{R}(w)}{\partial w} = -\varphi^{(n)} d^{(n)}$$

Stopping criterion = satisfactory classification rate.

Perceptron

Learning algorithm: 2-class case

1- Initialize randomly W_0

2- Randomly choose a couple of data $\left(x^{(n)}, d^{(n)}\right)$

3- Compute the output of the perceptron and compare it to $d^{(n)}$

$$f\left(w_0 + \sum_{i=1}^{D} w_i\, \varphi_i(x)\right) = f\left(w^T \varphi^{(n)}\right)$$

4- Adapt the weights:

If $x^{(n)}$ is well classified: $f\left(w^T \varphi^{(n)}\right) = d^{(n)}$

$$\longrightarrow w(t+1) = w(t)$$

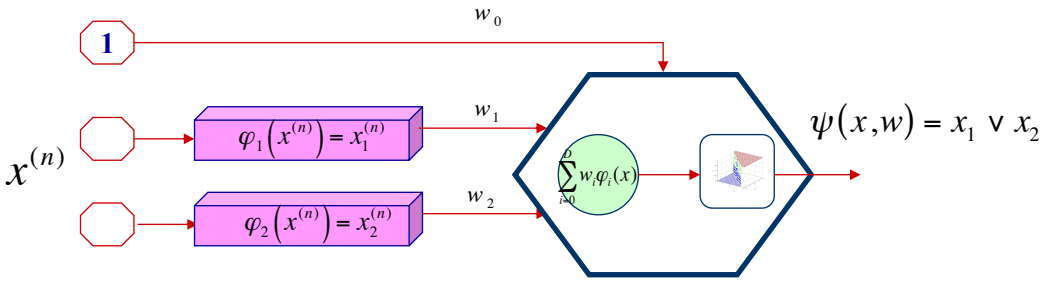
If $x^{(n)}$ is misclassified: $f\left(w^T \varphi^{(n)}\right) \neq d^{(n)}$

$$\longrightarrow w(t+1) = w(t) + \varepsilon(t)\, \varphi^{(n)} d^{(n)}$$

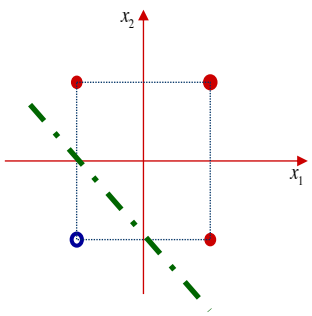
Where $\varepsilon(t)$ is de step of the gradient.

5- Repeat from 2 to 4 until an acceptable error value is obtained.

Perceptron example

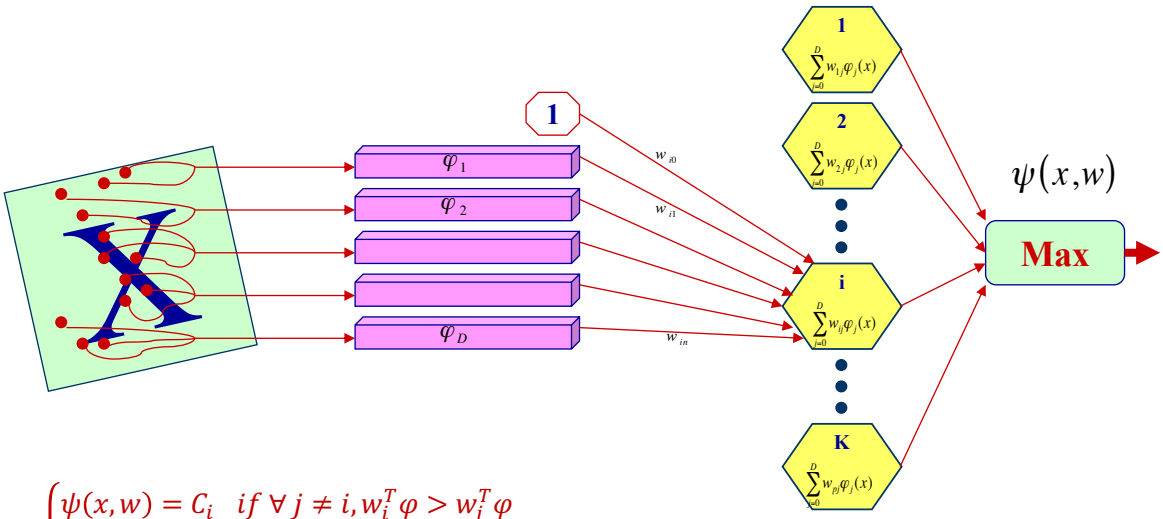


	x_0	x_1	x_2	d	
x^1	1	1	1	1	●
x^2	1	1	-1	1	●
x^3	1	-1	1	1	●
x^4	1	-1	-1	-1	○



Perceptron

p-class case: $C_1, C_2, \ldots, C_p$



$$\begin{cases} \psi(x,w) = C_i \quad if\ \forall j \neq i, w_i^T \varphi > w_j^T \varphi \\ \\ or \\ \\ \psi(x,w) = C_i \quad if\ \forall j \neq i, \sum_{k=1}^{D} w_{ik}\,\varphi_k > \sum_{k=1}^{D} w_{jk}\,\varphi_k \end{cases}$$

Perceptron

Learning algorithm: p-class case

1- Initialize randomly W_0

2- Randomly choose a learning example $x^{(n)} \in C_i$

3- Compute the output of the perceptron and compare it to C_i

$$\psi(x^{(n)}, w) = C_j \Leftrightarrow j = \underset{i}{\text{Argmax}}\left(w_i^T \varphi^{(n)}\right)$$

4- Adapt the weights:

If $x^{(n)}$ is well classified: $C_j = C_i$

$$\longrightarrow w(t+1) = w(t)$$

If $x^{(n)}$ is misclassified: $C_j \neq C_i$

$$\longrightarrow w_i(t+1) = w_i(t) + \varepsilon(t)\,\varphi^{(n)} d^{(n)}$$

$$\longrightarrow w_j(t+1) = w_j(t) - \varepsilon(t)\,\varphi^{(n)} d^{(n)}$$

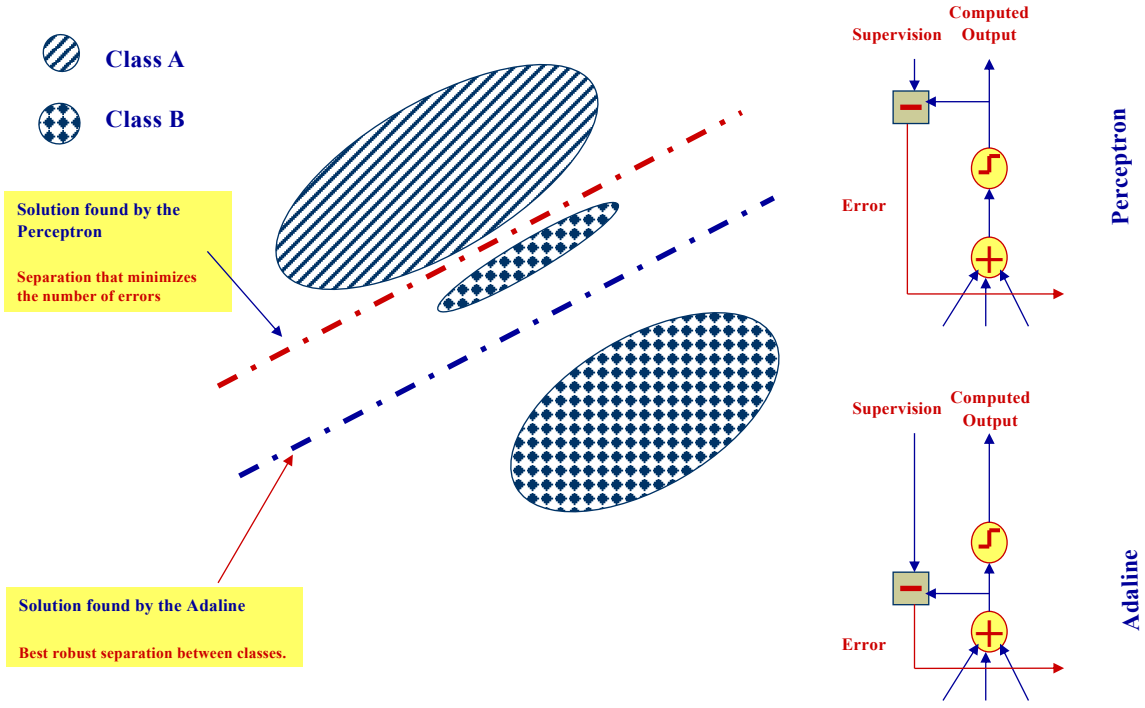
$$\longrightarrow w_k(t+1) = w_k(t)\ \forall\, k \neq i, j$$

Where $\varepsilon(t)$ is de step of the gradient.

5- Repeat from 2 to 4 until an acceptable error value is obtained.

Perceptron vs Adaline

2-class case



Perceptron convergence theorem

« If a set of examples is linearly separable, then the algorithm of Perceptron learning converges to a correct solution in a finite number of iterations »

Arbib M.A. (1987) : « Brains, Machines, and Mathematics »
Berlin, Springer-Verlag.

Rosenblatt F. (1962) : « Principles of Neurodynamics »
N.Y., Spartan.

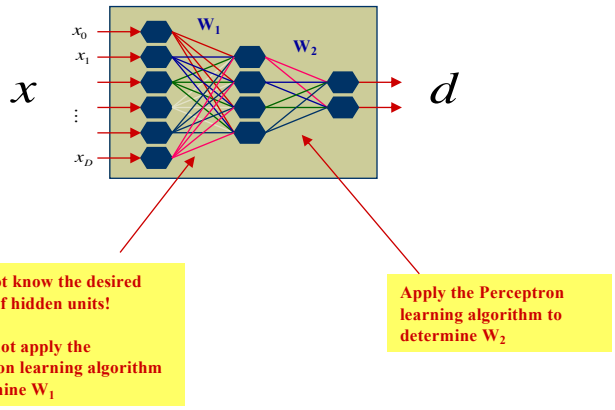
Block H.D. (1962) : « The Perceptron: A Model for Brain Functioning »
Reviews of Modern Physics 34, 123-135.

Minsky M.L. & Papert S.A. (1969) : « Perceptrons »
Cambridge, MIT Press.

Diederich S. & Opper M. (1987) : « Learning of Correlated Patterns
Spin-Glass Networks by Local Learning Rules »
Physical Review letters 58, 949-952.

Multi-layer Architecture

The credit assignment problem
Given a layered network, and a set of input-output pairs examples.



The Gradient Back - Propagation algorithm brings a simple solution to this problem.

The post connectionist break

Control issues

1963, Bryson A., Denham W., Dreyfuss S.,

"Gradient Back-propagation"

The 80s, the exact scope of the work of Minsky & Papert Will be correctly perceived.

The "rediscovery" of the "Gradient Back-Propagation"

1986, LeCun Y.

1986, Rumelhart D., Hinton G.E., Williams R.

Bryson A., Denham W., Dreyfuss S. (1963) : « Optimal Programming Problem With Ineduality Constraints. I: Necessary Conditions for Extremal Solutions », AIAA Journal, Vol. 1, pp. 25-44.

LeCun Y. (1986) : « Learning Processes in Asymmetric Threshold Network » Disordered Systems and Biological Organizations, Les Houches, France, Springer, pp. 223-240.

Rumelhart D., Hinton G.E., Williams R. (1986) : « Learning Internal Representations by Error Propagation » In Parallel Distributed Processing: exploring the microstructure of cognition, Vol I, Badford Books, Cambridge, MA, pp. 318-362, MIT Press.